

Vocabulary Insertion in two steps: Evidence from anti-homophony effects in Ethiosemitic*

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Abstract. Certain Amharic verbal constructions include a main verb and an auxiliary that are both inflected for subject agreement. However, in certain φ -feature combinations, one of the agreement markers is absent. We distinguish two patterns of agreement disappearance and develop an analysis of both on which full agreement in the syntax can be obscured during postsyntactic Vocabulary Insertion. To that end, we propose that Vocabulary Insertion be broken down into two ordered steps. First, *Select* chooses the best exponent to realize the features of a given node. Second, *Link* pairs the node and selected exponent. We associate the two types of agreement disappearance in Amharic with each of these steps: main verb agreement is realized as zero during Select via blocking, whereas auxiliary agreement goes unrealized due to a failure of Link. The failure of Link is driven by a novel anti-homophony constraint we dub Uniqueness. Finally, we show that our analysis accounts for cross-Ethiosemitic variation in the realization of agreement and is more empirically successful than alternatives.

Keywords. Vocabulary Insertion; anti-homophony; Uniqueness; multiple agreement; post-syntactic morphology; Amharic

*Thanks to Mengistu Amberber, Johanna Benz, Hagit Borer, Radwa Fathi, Laura Kalin, Ronny Meyer, Asia Pietraszko, Paul Portner, Jeff Punske, Chris Reintges, Aaron Rubin, Hannah Sande, and Elizabeth Zsiga, as well as attendees at NACAL 46, PLC 43, the Workshop on Perspectives on Templatic Morphology, ExoWords, WOMP 2024, ACAL 55, SLE 57, and AfAf2. Giant thanks to Meriem Tikue and Mengistu Amberber for consulting on the Amharic data (certain examples without a citation are from elicitation conducted by Kramer).

1 Introduction

In Amharic, a Semitic language spoken in Ethiopia, *complex verbs* (also known as *compound verbs*) are periphrastic tense constructions which consist of a main verb and an encliticized non-past auxiliary *-allä*. For example, the complex gerund in (2), combines a simple gerund like (1) with *-allä* and has a present perfect meaning.¹

- (1) säbrä-**a** (> *säbra*) (GERUND)
 break.GER-**3FSG**
 ‘her having broken’ (adapted from Leslau 1995: 355)
- (2) säbrä-**a** -allä-**ätŋŋ** (> *säbrallätŋŋ*) (COMPLEX GERUND)
 break.GER-**3FSG** -AUX.NPST-**3FSG**
 ‘she has broken’ (adapted from Leslau 1995: 387)

As (2) illustrates, both the main verb and the auxiliary can bear their own subject φ -agreement morphemes: the gerund bears the third feminine singular suffix *-a* while the auxiliary bears the third feminine singular suffix *-ätŋŋ*. This duplicate subject agreement marking is also attested with complex imperfectives, where main verb agreement consists of a prefix and a suffix. For instance, the complex imperfective verb in (4) combines the simple imperfective verb in (3), which bears the second person *ti-* prefix and the second feminine singular suffix *-i*, with the auxiliary *-allä*, which bears its own second feminine singular suffix *-f*. Complex imperfectives are interpreted as present (progressive) or future.

- (3) **ti-säbr-i** (> *tisäbri*) (SIMPLE IPFV)
2-break.IPFV-**2FSG**
 ‘you (F.SG) break’ (Leslau 1995: 301)

¹All Ethiosemitic glossed verbs are given in their underlying forms unless otherwise specified, with surface forms appearing in parentheses and to the right of ‘>’. Abbreviations follow the Leipzig Glossing Rules, with the following exceptions: A – default vowel in Kilega, FV – final vowel in Bantu verbs, GER – gerund, NPST – non-past.

- (4) **ti-säbr-i** -allä-**f** (> *tisäbr^yalläf*) (COMPLEX IPFV)
2-break.IPFV-2FSG -AUX.NPST-**2FSG**
 ‘you (F.SG) {break / will break}’ (adapted from Leslau 1995: 342)

Thus, multiple subject agreement marking is possible in Amharic complex verbs.

But that is not the whole story. In certain subject φ -feature combinations, expected agreement morphemes go missing. Consider the complex gerund in (5): while the gerund bears the second feminine singular suffix -*f*, the same suffix does not appear on the auxiliary, though it is expected to (compare (4), where -*allä* *does* bear -*f*).

- (5) **säbrä-f** -allä (*-**f**) (> *säbräfall*) (COMPLEX GERUND)
break.GER-2FSG -AUX.NPST (*-**2FSG**)
 ‘you (F.SG) have broken’ (adapted from Leslau 1995: 387)

Similarly, certain main verb agreement suffixes are missing in the complex imperfective. While the simple imperfective verb in (6) bears both a third person prefix *yi-* and a plural suffix -*u*, the -*u* suffix is absent from the main verb in the complex imperfective in (7).

- (6) **yi-säbr-u** (> *yiäbru*) (SIMPLE IPFV)
3-break.IPFV-PL
 ‘they break’ (adapted from Leslau 1995: 301)
- (7) **yi-säbr** (*-**u**) -allä-**u** (> *yiäbrallu*) (COMPLEX IPFV)
3-break.IPFV (*-PL) -AUX.NPST-**PL**
 ‘they {break / will break}’ (adapted from Leslau 1995: 342)

In this article, we develop a novel morphological account of missing agreement in Amharic complex verbs. On our analysis, there is consistently full agreement on auxiliaries and main verbs in the syntax (i.e. ‘(Agr-)V-Agr-AUX-Agr’). It is in the postsyntactic component that agreement affixes disappear. We posit two distinct postsyntactic mechanisms to govern the (dis)appearance of these affixes. Main verb agreement ‘disappears,’ as in (7), as a result of contextual allomorphy inserting a - $\emptyset$ suffix. By contrast, to explain the disappearance of auxiliary agreement, we propose a constraint dubbed “Uniqueness” which prohibits repeat insertions of the same exponent within a single M(orphological)-

word during Vocabulary Insertion, acting as an anti-homophony constraint. Uniqueness, we contend, is responsible for the disappearance of auxiliary agreement in the complex gerund in (5), as it would have been homophonous with main verb agreement—namely, *-f*. As a result, no exponent whatsoever is inserted to realize auxiliary agreement.

Providing an empirically adequate account of this constraint—including when it applies and how it interacts with other postsyntactic mechanisms—reveals deeper properties of how Vocabulary Insertion works and of the domains over which morphological constraints and operations apply. Because Uniqueness does not force a retreat to an elsewhere exponent but instead results in the absence of an exponent altogether, we propose that Vocabulary Insertion must be broken down into two ordered steps (see also Kalin 2022: esp. sec. 7.3).

The first step, dubbed *Select*, is responsible for selecting, for a given terminal node, which vocabulary entry maximally realizes the features of that node. The second step, dubbed *Link*, pairs the terminal node with the chosen vocabulary entry’s exponent. Uniqueness crucially applies to the second step:

(8) **The Uniqueness constraint (on Link)**

In an M-word, the exponent associated with a vocabulary entry α is Linked no more than once.

We contend that the effects of Uniqueness are mediated by its interaction with, among other things, bottom-up Vocabulary Insertion (Bobaljik 2000; Embick 2010; Kalin 2022), the formation of M-words (Embick and Noyer 2001; Arregi and Nevins 2012), and contextual allomorphy determined by linear adjacency (e.g. Embick 2010; Arregi and Nevins 2012; Merchant 2015; Ostrove 2018; Hewett 2023a,b; Kramer 2023: sec. 4). Thus, from a very specific constraint in a particular language, we tease out broader consequences for the syntax-morphology interface, in particular fleshing out the nature of Vocabulary Insertion.

At the same time, although we show that Uniqueness is active in Amharic, we also show that it is inactive in closely related languages whose complex verbs tolerate homophonous

agreement (as discussed in Bulakh 2014). This finding supports the conclusion that, while Uniqueness is a learnable constraint that is highly general in its application (insofar as it applies across complex paradigms), it is not a necessary property of Vocabulary Insertion or of morphological systems. Our analysis thus likens Uniqueness to other postsyntactic constraints which vary across languages and which motivate patterns of syncretism (via impoverishment), morphological discontinuities (via fission), and morpheme displacement.

The rest of this paper is organized into the following sections. Section 2 presents the basic empirical puzzle to be accounted for: disappearing auxiliary agreement in Amharic complex gerunds. Section 3 then develops an analysis of this disappearing auxiliary agreement based on a two-step Vocabulary Insertion process and the Uniqueness constraint. Section 4 motivates the M-word as the domain over which Uniqueness applies by demonstrating that homophonous agreement ‘reappears’ when the main verb and auxiliary occupy distinct complex heads. In section 5, we marshal an additional argument that homophony is what determines the disappearance of auxiliary agreement: allomorphy of main verb agreement bleeds Uniqueness and licenses the ‘reappearance’ of auxiliary agreement. Section 6 provides an interim summary of our findings. Then, in section 7, we argue that variation across Ethiosemitic in the realization of agreement is accounted for if Uniqueness is a learned constraint, rather than being a fundamental property of Vocabulary Insertion. In section 8, we show that our Uniqueness-based analysis is empirically superior to various alternatives. Section 9 concludes and considers broader implications of our analysis.

2 Disappearing AUX agreement in Amharic complex gerunds

We begin by describing the distribution of agreement affixes with simple and complex gerunds in Amharic. In particular, we motivate the following generalization characterizing the (dis)appearance of auxiliary agreement:

(9) AUX AGREEMENT GENERALIZATION IN AMHARIC COMPLEX VERBS

Auxiliary agreement disappears whenever it would be homophonous with main verb agreement.

We will account for this generalization with the Uniqueness constraint in Section 3.

Amharic gerunds (also called ‘converbs’) appear in subordinate, typically adverbial, clauses and are interpreted as aspectually perfective or temporally anterior relative to the predicate their clause modifies (Leslau 1995: sec. 59.2; Meyer 2016a: 184–187). Take (10), for example: the gerund ‘insult’ heads a clause modifying the main verbal predicate ‘I hit him,’ and the gerund clause is interpreted as a causal adjunct.

- (10) [sädbä-^w-ññ] mätta-hu-t. (GERUND)
 [insult.GER-3MSG-1SG.O] hit.PFV-1SG-3MSG.O
 ‘I hit him [because he insulted me].’ (adapted from Leslau 1995: 361)

Gerunds bear an agreement suffix indexing the φ -features of the subject of the gerund clause;² in (10), this suffix is a floating labial feature -^w realizing third person masculine singular subject agreement.³ The full paradigm of gerund subject φ -agreement is given in Table 1, with agreement affixes in boldface.

	SG	PL
1	säbirr- e	säbrä- n
2M	säbrä- h	säbrä- afǧǧihu
2F	säbrä- ǰ	säbrä- afǧǧihu
3M	säbrä- ^w	säbrä- u
3F	säbrä- a	säbrä- u

Table 1: Gerund of *säbbärä* ‘break’ (Leslau 1995: 355)

²The gerund in (10) also bears a first person singular object marker -ññ, but this is orthogonal to our primary focus, which is the form of subject agreement markers. For detailed discussion of object markers in Amharic, see Kramer (2014) and Baker and Kramer (2018).

³This floating labial feature gets realized by docking onto the preceding stem-final vowel and changing that vowel’s quality, e.g. underlying *säbrä-^w* is realized as *säbro*.

As can be seen in this table, there is a distinct exponent of subject φ -agreement for each combination of person and gender features in the singular and there is a distinct exponent for each person in the plural.

The simple gerund can be combined with the enclitic non-past auxiliary *-allä*⁴ to yield a complex gerund, as in (11).

- (11) *yä-abbat-e-n* *wäre sämitf-tf-e* *-allä-h^w*. (COMPLEX GERUND)
 of-father-1SG-ACC news hear.GER-1SG -AUX.NPST-1SG
 ‘I have heard the news of my father.’ (adapted from Leslau 1995: 389)

Like the simple gerund, the complex gerund is interpreted as present perfect, though it is not restricted to subordinate clauses (Leslau 1995: sec. 60.2). As (11) illustrates, the auxiliary *-allä* can bear its own subject φ -agreement suffix (here, *-h^w*) in addition to that which appears on the gerund. The full agreement paradigm for *-allä* is given in Table 2.⁶

Given that multiple exponence of subject φ -agreement in the complex gerund is possible in principle, as in (11), we might expect it to persist in all forms of the complex gerund. But this is not the case: auxiliary agreement is missing in all cells except 1SG, 3MSG, and 3FSG. This is shown for the complex gerund of the verb *säbbärä* ‘break’ in Table 3; for

⁴We take the auxiliary stem to include a final *-ä* vowel (see Kramer To appear and references cited there).

⁵See Leslau (1995: 387, 528) for other surface variants of the first singular auxiliary agreement suffix.

⁶The surface forms of the auxiliary often differ from the underlying forms in Table 2. For instance, *-allä* with 3MSG agreement appears on the surface simply as *-all* (Leslau 1995: 387). We derive this surface form from underlying *-allä-ä* via a sequence of two (morpho)phonological operations. First, when two identical vowels meet in hiatus in Amharic, one of them is deleted (Leslau 1995: 35–36). Second, word-final *-ä* is generally truncated on enclitic *-allä* (Leslau 1995: 387, 527; though Leslau 1995: 387, fn. 1 reports regional variation in this regard). The derivation in (i) demonstrates how these processes yield *-all* from *-allä-ä*:

- (i) a. UNDERLYING FORM: *-allä-ä*
 b. VOWEL HIATUS RESOLUTION: *-allä*
 c. TRUNCATION: *-all*

Thus, the surface form of auxiliary agreement in Amharic is predictable from regular (morpho)phonology.

	SG	PL
1	-allä- h^w	-allä- n
2M	-allä- h	-allä- atʃʃih^w
2F	-allä- ʃ	-allä- atʃʃih^w
3M	-allä- ä	-allä- u
3F	-allä- ätʃʃ	-allä- u

Table 2: Non-past enclitic auxiliary agreement (adapted from Leslau 1995: 342)

readability, the auxiliary complex (i.e. auxiliary root and auxiliary agreement) is enclosed in square brackets and boxes indicate where agreement is expected but missing.

	SG	PL
1	säbirr- e^y [-allä- h^w]	säbrä- n [-allä]
2M	säbrä- h [-allä]	säbrä- atʃʃih^w [-allä]
2F	säbrä- ʃ [-allä]	säbrä- atʃʃih^w [-allä]
3M	säbrä- w [-allä- ä]	säbrä- w [-allä]
3F	säbrä- a [-allä- ätʃʃ]	säbrä- w [-allä]

Table 3: Complex gerund of *säbbärä* ‘break’ (adapted from Leslau 1995: 387)

While the gaps in Table 3 may initially seem haphazard, we contend that the (dis)appearance of auxiliary agreement is crucially linked to the form of subject agreement markers. Building on Diertani and Eilam (2010), we propose the following descriptive generalization:⁷

- (12) **AUX AGREEMENT GENERALIZATION IN AMHARIC COMPLEX GERUNDS**
 Auxiliary agreement disappears whenever it would be homophonous with main verb agreement.

As the reader can verify by comparing Tables 2 and 3, auxiliary agreement is only manifested overtly in the complex gerund when it is formally distinct from gerund agreement.⁸ But this raises the question: why should homophony affect the (dis)appearance of auxiliary

⁷It is unlikely that the disappearance of the agreement suffixes is due to phonological factors, since the missing suffixes do not form a phonological natural class. See also Diertani and Eilam (2010: 4).

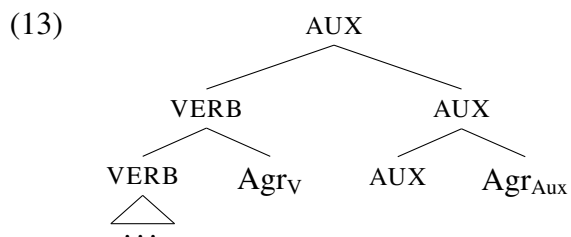
⁸The third plural *-w* gerund agreement suffix and third plural *-u* auxiliary agreement suffix are identical.

agreement? Answering this question is the aim of the next section.

3 Uniqueness at work: anti-homophony at Vocabulary Insertion

In a nutshell, we propose that auxiliary agreement in Amharic fails to be realized when the following conditions hold: (i) the exponent selected to realize auxiliary agreement is homophonous with main verb agreement (see (12)); and (ii) the main verb and auxiliary form a single complex head. In particular, we argue that auxiliary agreement ‘disappears’ when Vocabulary Insertion—the operation in Distributed Morphology (DM) (Halle and Marantz 1993, 1994) which assigns exponents to syntactic terminal nodes—fails to insert an exponent, for reasons to be elaborated in Section 3.2.

We furthermore offer a derivational ordering explanation as to why auxiliary agreement and not main verb agreement disappears when the two are homophonous. Suppose that complex gerunds involve a structure along the lines of (13) (to be fleshed out and justified in Section 3.1), where the main verb and auxiliary form a single complex head and each bears its own agreement suffix, represented here with Agr_V and Agr_{Aux} , respectively.



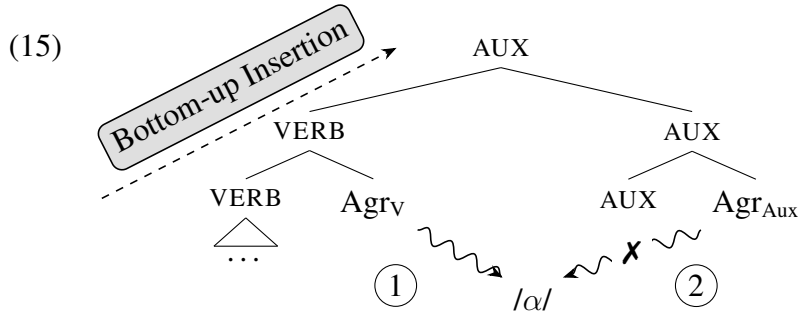
Adopting the claim from prior work that Vocabulary Insertion operates cyclically from the

It is common in Amharic for the same morpheme to alternate between being a vowel and being a glide in different contexts (for example, the definite marker also alternates between *-u* and *-w* (Leslau 1995: 155)).

A reader who is familiar with Amharic might wonder then why the third plural auxiliary surfaces as *all-u* when used as the copula, rather than surfacing as *allä-w*. We assume that the final vowel of *allä* deletes when followed by another vowel (leading to *all-u*) and when it is word-final. See fn. 6.

bottom up (e.g. Bobaljik 2000; Embick 2010; Kalin 2022), we account for the asymmetric disappearance of auxiliary agreement as follows: main verb agreement, which is lower in the complex head than auxiliary agreement, will be inserted first, bleeding insertion of homophonous auxiliary agreement. The derivation in (15) illustrates this: if an earlier cycle of Vocabulary Insertion realizes Agr_V with the vocabulary entry in (14)—where realization is represented via squiggle arrows connecting the terminal node with the output of the rule—then a later cycle of Vocabulary Insertion cannot realize Agr_{Aux} with the same rule.

(14) $\text{Agr} \rightarrow /a/$



To account for this anti-homophony pressure and to derive the derivational bleeding effect, we propose a novel, exponence-based constraint on Vocabulary Insertion dubbed the *Uniqueness* constraint in Section 3.2. Before laying out our proposal, however, we first set the stage on Amharic clause structure in Section 3.1. Then, in Section 3.3, we demonstrate how the particular combination of our claims about Amharic clause structure and the Uniqueness constraint, together with bottom-up Vocabulary Insertion, derive the distribution of agreement markers in the complex gerund.

3.1 Background on Amharic clause structure and complex heads

Previous work on Amharic clause structure has delivered the following important results (for additional work in this vein beyond the sources cited in this section, see Girma Halefom 1994; Baker 2012; Kramer 2014, 2023, To appear; Mulusew Asratie Wondem 2014; Leung

and Girma Halefom 2017; Desalegn Workneh 2020, 2021, 2022).

First, main verb agreement has been argued to be located on (high) Aspect (Girma Awgichew Demeke 2003: 23). This accounts for aspect-sensitive subject agreement allomorphy, exemplified for instance by the contrast between perfective ((16a)) and gerund ((16b)) second masculine singular agreement (we follow Girma Awgichew Demeke 2003: 180–182 in taking perfective, imperfective, and gerund to be flavors of Asp).

- (16) *Subject agreement displays aspect-sensitive allomorphy in Amharic*
- | | | |
|----|--|----------|
| a. | säbbär- k
break.PFV- 2MSG
‘you (M.SG) broke’ (Leslau 1995: 287) | (PFV) |
| b. | säbrä- h
break.GER- 2MSG
‘you (M.SG) having broken’ (Leslau 1995: 355) | (GERUND) |

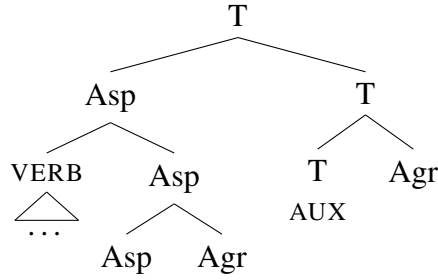
Following Chomsky (2000, 2001) *et seq.*, this means that Asp bears a φ -probe.

Second, auxiliaries like *-allä* are arguably located in T in Amharic (Girma Awgichew Demeke 2003; Baye Yimam 2006) because they correlate with tense distinctions (see Goldberg 1964; Meyer 2016a,b). We propose that auxiliaries independently probe the subject DP for φ -features. Thus, there is full agreement on both T and Asp in the syntax.⁹ We represent T/Asp agreement as Agr nodes adjoined to T/Asp since the realization of agreement is distinct from the realization of T (the auxiliary *-allä*) and from the realization of Asp (root-and-pattern morphology, compare (16a) to (16b)).

Third, the verb and associated functional heads (e.g. Asp, T) typically combine to form a complex head (Baker and Kramer 2014), along the lines of (17).

⁹The fact that multiple heads in the clause agree with the same (subject) DP is reminiscent of ‘hyperagreement’ patterns in Bantu compound tense constructions (on which see e.g. Carstens 2001, 2011; Henderson 2006; for similar patterns in other languages, see Deal To appear: sec. 5). The existence of hyperagreement may suggest that the Activity Condition of Chomsky (2000: 127ff., 2001: 6ff.) is inactive in Amharic.

(17) *Basic structure of Amharic complex verbal heads* (see Diertani 2011: 47–52)



There are various indications that the main verb and non-past auxiliary *-allä* comprise a single complex head in Amharic. First, freestanding words do not intervene between the main verb and this auxiliary. Second the main verb and auxiliary form a single domain for stress assignment (Stolen 2013: 12; Sande and Hedding 2017: 73). Finally, *-allä* is written as a single unit with the main verb, paralleling the orthographic behavior of other verbal enclitics and suffixes (see Kramer To appear: 4). For the most part, it will not matter to our analysis how exactly the complex head in (17) is formed (raising/lowering, syntactically/post-syntactically, etc.); however, see Section 4 for some initial evidence that the verb and T might combine post-syntactically.

Having laid out our assumptions about Amharic clause structure and complex heads, we now turn to our analysis of missing auxiliary agreement in the complex gerund.

3.2 The Uniqueness Constraint

Consider the following description of Vocabulary Insertion from Embick (2010: 12):

- (18) “[Vocabulary Insertion] is a procedure by which morphemes in a syntactic structure are assigned a phonological form [...] [I]ndividual *Vocabulary items* [...] compete for insertion at a given node, and the most specific one that can apply gives that node its phonological matrix.”

In this passage, Embick identifies two essential functions for Vocabulary Insertion. Its first function is to choose the best rule to realize the features of a given node. Its second function is to realize this node with the exponent associated with the chosen rule. This

characterization of Vocabulary Insertion is typical within the DM literature (see e.g. Arregi and Nevins 2012; Kramer et al. To appear: sec. 3.2).

We propose that these two functions are in fact autonomous and that each is associated with its own (sub-)operation that applies step-wise during Vocabulary Insertion (see Kalin 2022: esp. sec. 7.3 for an important precedent). (19) summarizes our proposal:

(19) **Vocabulary Insertion in two steps**

- a. *Select*: For a terminal node X^0 , choose the vocabulary entry α that maximally realizes the features of X^0 in accordance with the *Subset Principle* and the *Elsewhere Condition*.
- b. *Link*: Link the exponent associated with α to X^0 .

The first step of Vocabulary Insertion is to *select* the vocabulary entry that optimally matches the features and contextual specification of a given node. Whether or not a vocabulary entry matches a given node is determined by the Subset Principle (aka underspecification):

- (20) **Subset Principle** (adapted from Arregi and Nevins 2012: 117, (7))
Only a vocabulary entry whose morphosyntactic specification matches the morphosyntactic specification of a terminal node X^0 and whose contextual restriction matches the context of X^0 is eligible for Vocabulary Insertion at X^0 .

Competition among multiple eligible vocabulary entries is regulated by the Elsewhere Condition in (21), which relies on the definition of specificity in (22).

- (21) **Elsewhere Condition** (see Kiparsky 1973: esp. 94, (4))
If more than one vocabulary entry can apply to a terminal node, choose the most specific vocabulary entry.
- (22) *Specificity among vocabulary entries* (see Arregi and Nevins 2012: 117, (8))
Vocabulary entry α is more specific than vocabulary entry β if the set of contexts in which α can apply are a proper subset of the set of contexts in which β can apply.

Once a vocabulary entry has been selected, the second step is to *link* the exponent of the chosen entry with the terminal node. *Linking* is tantamount to *inserting*, except that Link explicitly modifies the representation by adding a squiggle arrow to connect a terminal

node with an exponent. Example (15) schematically illustrates the output of Link.¹⁰

By distinguishing Select from Link, we can posit independent constraints on each sub-operation. This is the launching point for our analysis of missing auxiliary agreement in Amharic: we posit the Uniqueness constraint in (23), which applies to Link but not to Select and which relies on the definition of M-word in (24).

(23) **The Uniqueness constraint (on Link)**

In an M-word, the exponent associated with a vocabulary entry α is Linked no more than once.

(24) **M-word** (= morphological/morphosyntactic word)

An M-word is a 0-level node that is not dominated by any other 0-level node. (Arregi and Nevins 2012: 239, (4); see also Embick and Noyer 2001: 571, (33))

Uniqueness forestalls the realization of iterative, homophonous agreement markers in a complex head. Importantly, because Uniqueness applies to Link only, it will not prevent a vocabulary entry from being *selected* multiple times. We leverage this distinction to explain why failure to realize homophonous agreement leads to the *absence* of an exponent in Amharic, rather than the appearance of a less specified form (i.e. an elsewhere), as detailed in the next section.

3.3 Deriving agreement in the Amharic complex gerund

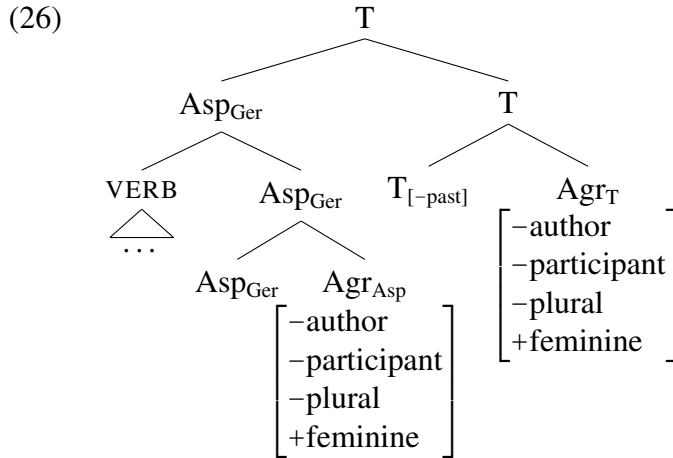
Let us now turn to deriving the distribution of agreement markers in the Amharic complex gerund. We begin with non-homophonous agreement, where Uniqueness does not apply.

¹⁰See also the results in Kalin and Rolle (2023), which are compatible with breaking Vocabulary Insertion down into multiple steps. Specifically, Kalin and Rolle propose that two types of subcategorization frames for exponents ought to be distinguished: Conditions on Insertion (i.e. ‘COINs’) are garden variety contextual restrictions on vocabulary entries, whereas Conditions on Position (i.e. ‘COPs’) determine the linear position of an exponent relative to a pivot. Under our approach, COINs would apply during Select, whereas COPs would be evaluated at least as late as Link, since COPs operate over strings of linked(/inserted) exponents.

For instance, in the third feminine singular complex gerund in (25), agreement on the main verb is suffixal *-a* while agreement on the non-past auxiliary is suffixal *-ätf*.

- (25) säbrä-**a** -allä-**ätf** (> säbrallätf) (COMPLEX GERUND)
 break.GER-**3FSG** -AUX.NPST-**3FSG**
 ‘she has broken’ (adapted from Leslau 1995: 387)

We adopt the following inventory of φ -features (see Hewett 2023b: 132, (4) and Kramer To appear: 3–4): [$\pm$ author, $\pm$ participant] for person (Noyer 1992; Halle 2000), [$\pm$ plural] for number (Halle 2000; Kramer 2016), and [$\pm$ feminine] for gender (Kramer 2015). Thus, the third feminine singular complex gerund in (25) can be represented as in (26).



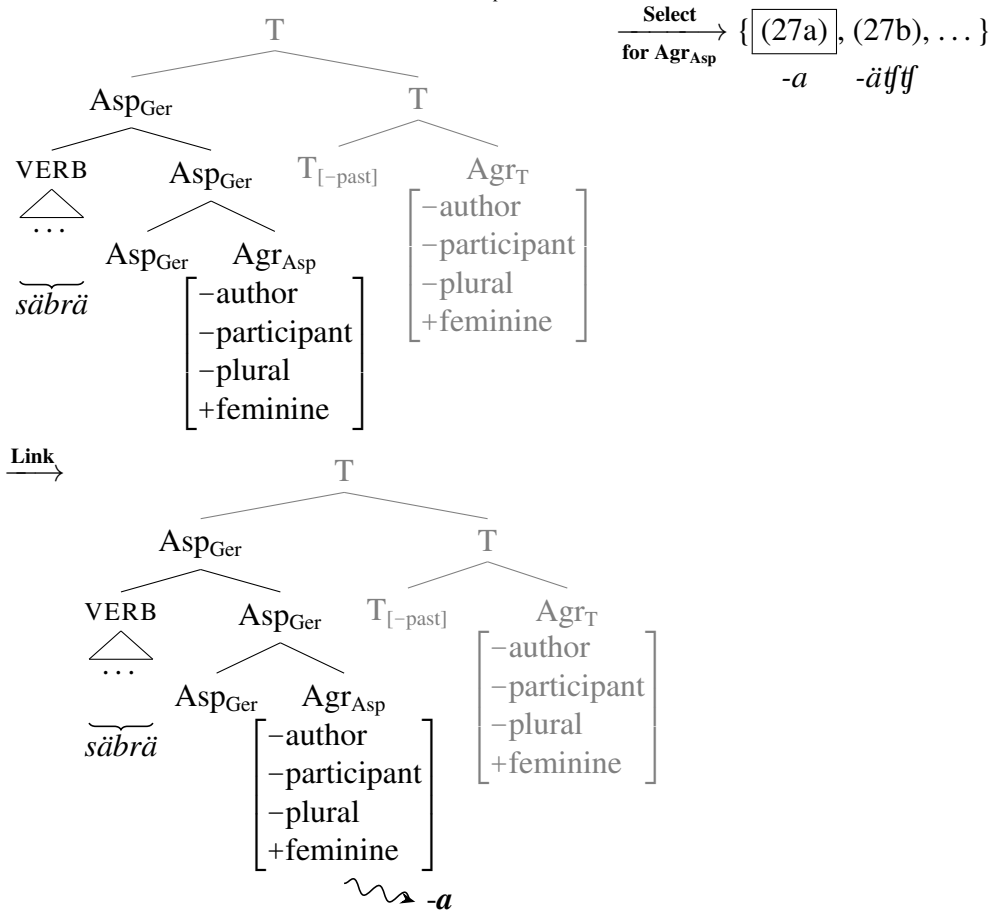
Focusing on agreement (and setting aside the verb, Asp, T, etc.), we posit the two vocabulary entries in (27) to realize a subset of the features on each of the Agr nodes. These two entries realize the same features, but (27a) also has a contextual restriction—namely, a left-adjacent Asp_{Ger} node. Following Hewett (2023a: 1111, fn. 16), we propose that the hierarchical arrangement and linear order of terminal nodes within the M-word are both fixed and legible to Vocabulary Insertion to account for such linearity-based allomorphy (see Hewett 2023a,b and Kramer 2023 on affixal allomorphy across Semitic), though we do not represent linear order information in our derivations for simplicity.

- (27) *Vocabulary entries for 3FSG agreement*

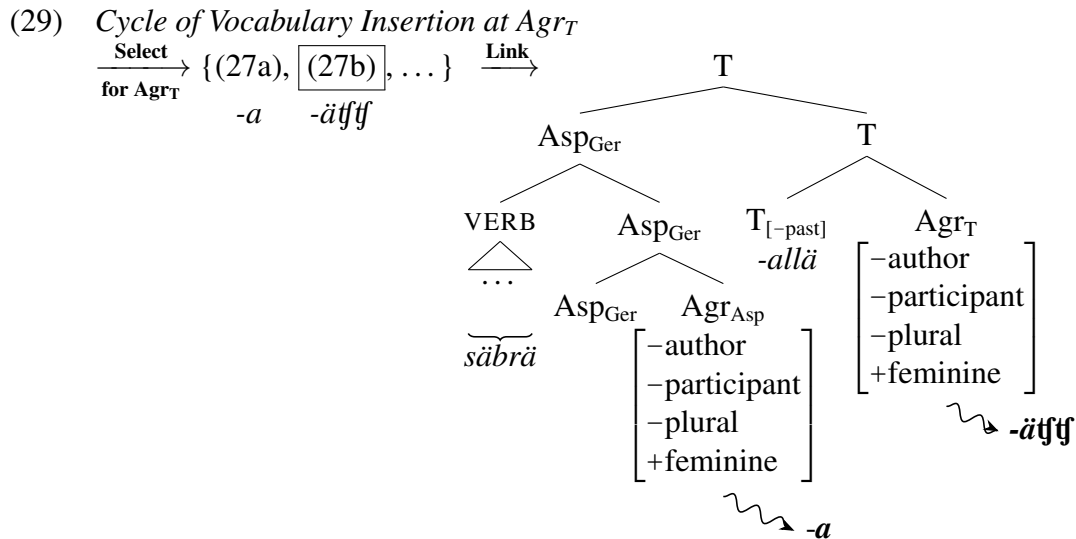
- a. $\begin{bmatrix} -\text{participant} \\ -\text{plural} \\ +\text{feminine} \end{bmatrix} \rightarrow -a / \text{Asp}_{\text{Ger}} \text{ ______}$ b. $\begin{bmatrix} -\text{participant} \\ -\text{plural} \\ +\text{feminine} \end{bmatrix} \rightarrow -\ddot{a}t\ddot{f}f$

Vocabulary Insertion will then apply node-by-node, selecting entries and linking their corresponding exponents, working its way from the bottom of the structure up. Accordingly, the first set of agreement features to be realized will be that on Asp. As a first step, Select determines which of the two vocabulary entries in (27) is the optimal candidate to realize the features of Agr_{Asp} . Although both rules could apply, (27a) is more specific, so it is chosen in accordance with the Elsewhere Condition ((21)). Then, Link links Agr_{Asp} with the exponent *-a*. This cycle of Vocabulary Insertion is shown in (28), where heads not yet targeted for Vocabulary Insertion are grayed out.

(28) *Cycle of Vocabulary Insertion at Agr_{Asp}*



Vocabulary Insertion will then apply to Agr_T. In this case, the only vocabulary entry that can apply is (27b), so this rule is chosen and its exponent is linked to Agr_T:



Because each exponent is linked only once within the complex head, the Uniqueness constraint does not apply.

By contrast, Uniqueness will apply to block the realization of higher homophonous auxiliary agreement, for instance in the second feminine singular complex gerund in (30).

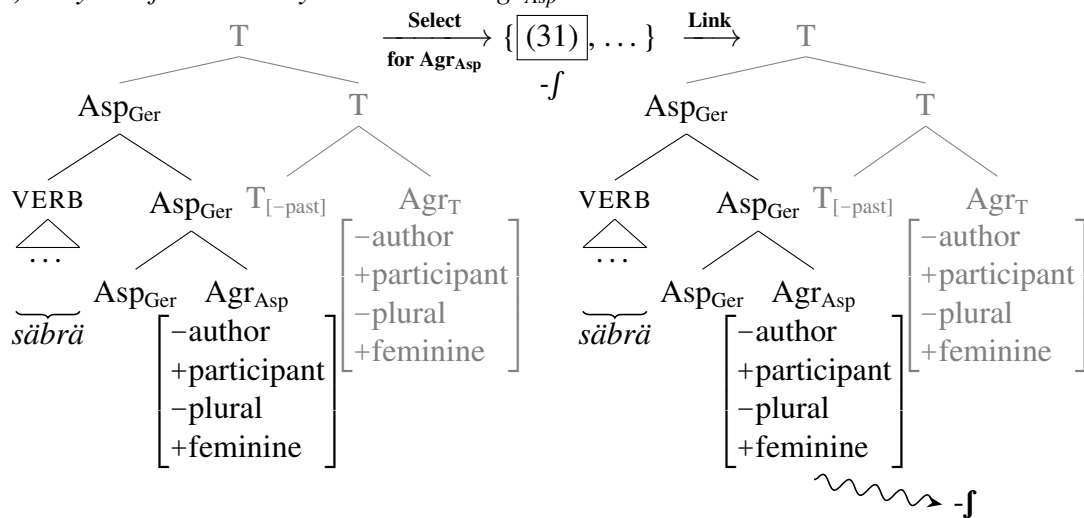
- (30) säbrä-**f** -allä (*-**f**) (> *säbräfall*) (COMPLEX GERUND)
 break.GER-**2FSG** -AUX.NPST (*-**2FSG**)
 ‘you (F.SG) have broken’ (adapted from Leslau 1995: 387)

We propose that there is just one vocabulary entry available to realize second feminine singular agreement:

- (31) *Vocabulary entry for 2FSG agreement*
- $\left[\begin{array}{l} +\text{participant} \\ -\text{plural} \\ +\text{feminine} \end{array} \right] \rightarrow -f$

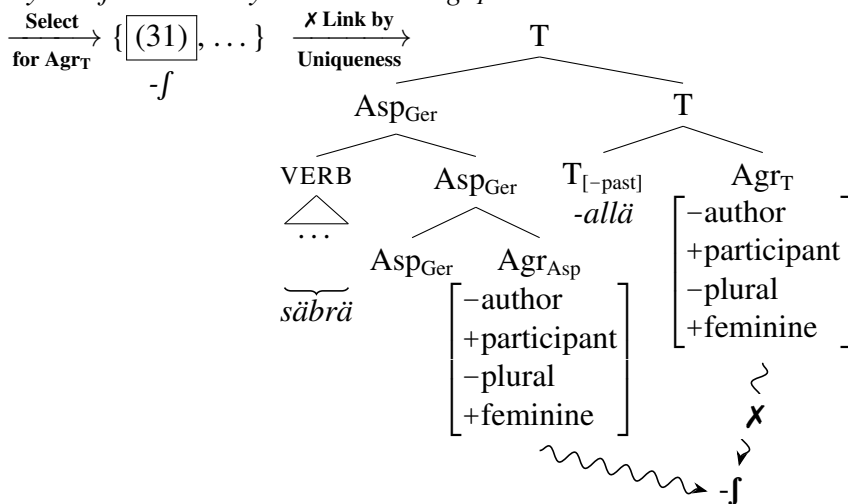
The cycle of Vocabulary Insertion over Asp will encounter no difficulties: (31) will be selected and its exponent -*f* linked to Agr_{Asp}.

(32) *Cycle of Vocabulary Insertion at Agr_{Asp}*



When Vocabulary Insertion reaches Agr_T , the same vocabulary entry ((31)) will be selected. However, any attempt to link $-f$ to Agr_T will induce a violation of the Uniqueness constraint, which says that an exponent may be linked no more than once within an M-word. We propose that Link fails in this scenario:¹¹

(33) *Cycle of Vocabulary Insertion at Agr_T*



¹¹Johanna Benz (*pers. comm.*) wonders why it is not possible to generate a second, duplicate $-f$ that Agr_T links to, thereby pathologically circumventing Uniqueness. To rule this out, we propose that the vocabulary entries evaluated during Vocabulary Insertion within an M-word form a set. It is thus impossible to generate a duplicate vocabulary entry, with its corresponding exponent $-f$, because, per set theory, $\{A, A\} = \{A\}$.

Importantly, failure of Link must not be fatal, since the derivation converges, just without linking the exponent $-f$ to Agr_T .

In fact, it must be the case that *no exponent* gets linked to Agr_T in this derivation. This finding is embodied in our generalization about auxiliary agreement in (12): when auxiliary agreement would be homophonous with main verb agreement, auxiliary agreement is absent. There is no retreat to an elsewhere form with failed Link.¹² To explain why auxiliary agreement is missing, rather than realized as an elsewhere, we propose that, in a given cycle of Vocabulary Insertion at node X^0 , Select and Link apply sequentially, in that order. Failure of Link does not allow Select to reevaluate the best entry for X^0 , excluding entries whose exponents cannot be linked.¹³ If Link fails, the derivation simply moves on.

To summarize, our account of the disappearance of auxiliary agreement in the Amharic complex gerund rests on three claims: (i) Vocabulary Insertion applies bottom up; (ii) Vocabulary Insertion applies in two steps (Select $\prec$ Link); and (iii) the Uniqueness constraint prohibits repeated linking of the same exponent within an M-word. Because Uniqueness applies within M-words, we expect it to fail to apply across M-words, with the result that homophonous agreement should resurface in such contexts. As we show in the next section, this prediction is borne out with Amharic complex gerunds inside relative clauses.

¹²This anti-homophony effect thus differs from the dissimilatory processes discussed in Nevins (2012: sec. 3.5), where an anti-homophony constraint feeds allomorph selection. Anti-homophony in Amharic also does not operate under (linear or structural) adjacency, unlike the cases described by Nevins (*ibid.*).

¹³It remains an open question how the grammar deals with a failure to Select an optimal vocabulary entry. One possibility is that *ineffability* results when no vocabulary entry is found which can realize the features of a given terminal node (see e.g. Kennedy and Merchant 2000, Arregi and Nevins 2014, Mendes and Nevins 2022, and Arregi and Hewett 2025: sec. 5).

4 No anti-homophony across M-words

In this section, we motivate our claim that Uniqueness applies within M-words. This claim provides further evidence for the M-word being a relevant domain for morphological operations and constraints, building on Embick and Noyer (2001); Embick (2007); Arregi and Nevins (2012, 2022); Pavlou (2018); Pak (2020); and Akkuş (2024), among others. In brief, we show that homophonous auxiliary agreement resurfaces when the main verb and auxiliary are separated into two M-words in Amharic, specifically in relative clauses.

Amharic relative clauses are prenominal and are headed by the complementizer *yä-*, which appears prefixed to the highest verbal element (i.e. verb or auxiliary) within the relative (Baker and Kramer 2014: 157–160).¹⁴ When the highest verbal element is the lexical verb, for instance in the perfective aspect, *yä-* prefixes to V:

- (34) [CP k’äyy mäkina **yä**-gäzz-a]-w astämari
 [CP red car C-buy.PFV-3MSG]-DEF teacher.M
 ‘the teacher who bought a red car’ (Baker and Kramer 2014: 159, (63a))

When the highest verbal element is an auxiliary like *-allä*, as in clauses with a complex gerund, *yä-* prefixes to the auxiliary. In such cases, homophonous auxiliary agreement strikingly reappears (see also Diertani and Eilam 2010: 8, (27)). Compare (35) with (36).

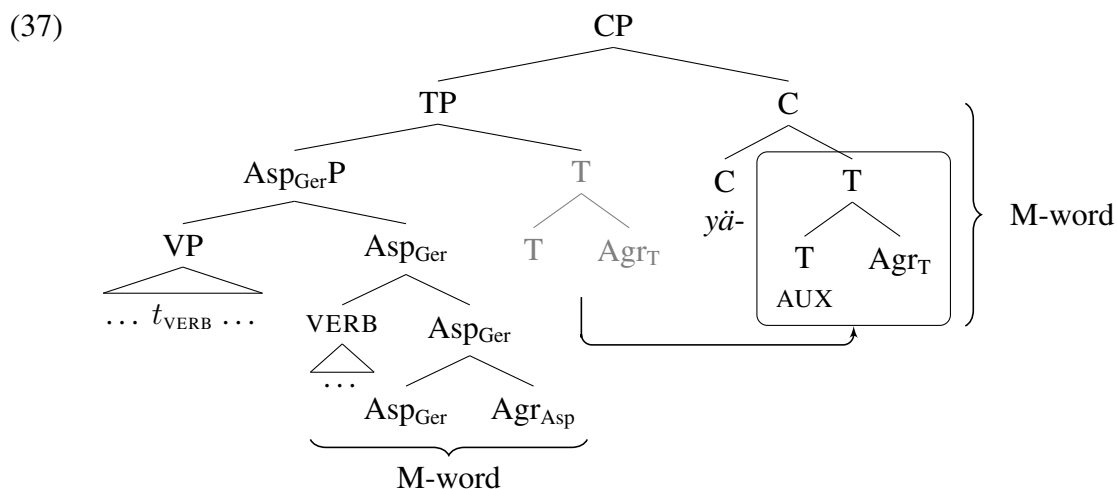
- (35) [CP k’ädmän dārsä-**n** y-allä-**n**]-äw iñña nän.
 [CP first arrive.GER-**1PL** C-AUX.NPST-**1PL**]-DEF we COP.1PL
 ‘It is us who arrived first.’ (Leslau 1995: 390)
- (36) dārsä-**n** -allä (*-**n**) (> *dārsänall*) (COMPLEX GERUND)
 arrive.GER-**1PL** -AUX.NPST (*-**1PL**)
 ‘we have arrived’

The 1PL auxiliary agreement suffix *-n* is homophonous with main verb agreement and is prohibited in the complex gerund in (36), where, as we argued in Section 3.1, the main verb

¹⁴Note that definite markers modifying the head N attach to the right edge of the relative clause. See Kramer (2010) for discussion and analysis.

and auxiliary form a single M-word. In (35), by contrast, where C is prefixed to *-allä*, *-n* appears on both the main verb and the auxiliary.¹⁵

We propose that auxiliary agreement ‘reappears’ because, in relatives, C and the auxiliary in T form a separate M-word from the main verb as a result of T-to-C movement. As (37) shows, after T and its associated agreement raise to C, C and T will form an M-word to the exclusion of the main verb in Asp.



The structure in (37) raises a question about the timing of head movement. On our analysis, T-to-C movement must precede and hence bleed the mechanism which normally combines T and V into a complex head in Amharic (compare (17)). This parallels the puzzle presented by English *do*-to-C movement bleeding T-to-V lowering. One way to ensure this timing in Amharic would be to propose that T-to-C movement in relative clauses is syntactic, while T and V are combined into a single complex head postsyntactically (see Embick and Noyer 2001: sec. 7.2 and Harizanov and Gribanova 2019: 481, fn. 23 for similar accounts of English T movement).

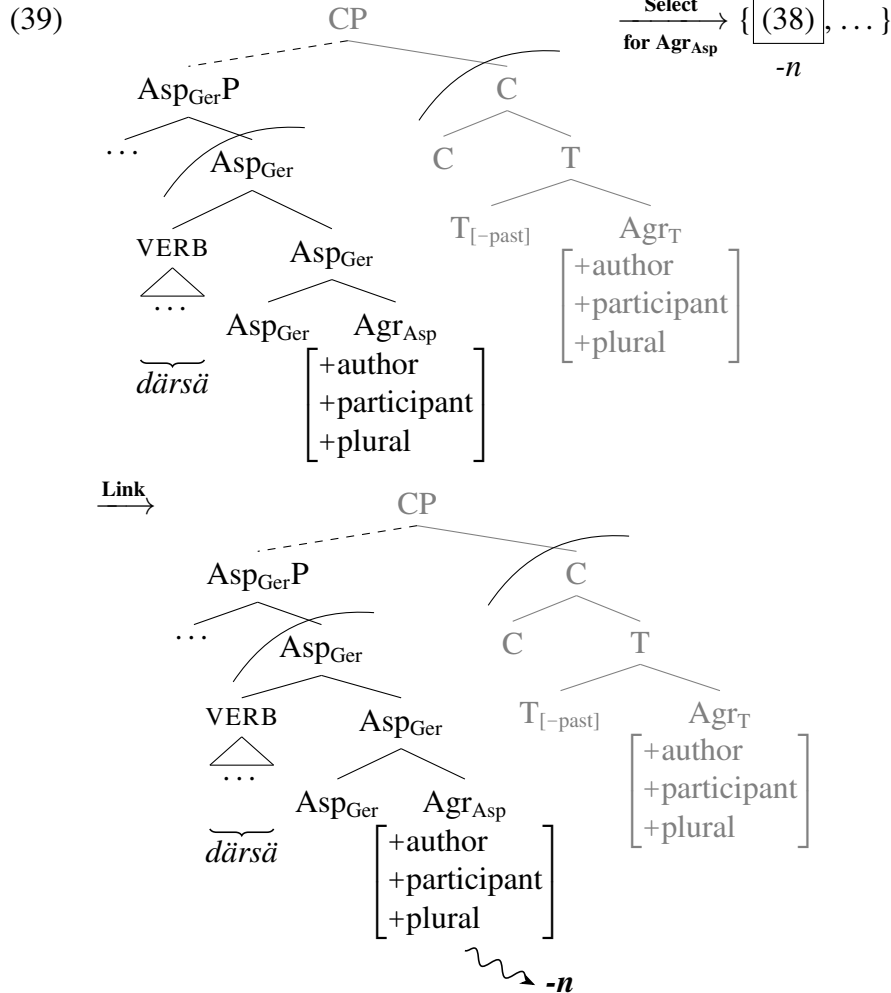
If (37) is the right structure for complex gerunds in relative clauses in Amharic, then our analysis straightforwardly explains the presence of homophonous agreement markers on V

¹⁵The complex gerund is illicit in any other kind of subordinate clause (Leslau 1995: 390).

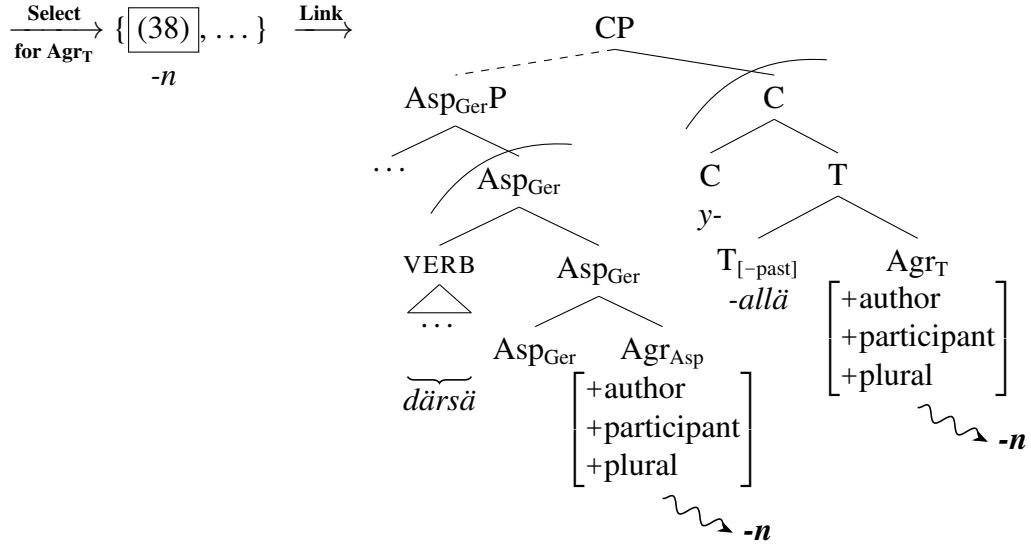
and AUX: the Uniqueness constraint does not block linking of homophonous agreement across M-words. As the derivation of (35) in (39) shows, the vocabulary entry in (38) can be selected and its exponent linked to an Agr node in both the Asp M-word and the C M-word without running afoul of Uniqueness.¹⁶ We use separate links and exponents for each M-word to illustrate that there is no problem for Uniqueness.

(38) *Vocabulary entry for 1PL agreement*

$$\begin{bmatrix} +\text{author} \\ +\text{participant} \\ +\text{plural} \end{bmatrix} \rightarrow -n$$



¹⁶Gender features are missing in (39) because gender is postsyntactically deleted from first person feature bundles prior to Vocabulary Insertion via *impoverishment* across Semitic, see Hewett (2023b: 145, (29)).



In summary, our analysis of disappearing auxiliary agreement in Amharic predicts that agreement should reappear when the main verb and auxiliary occupy distinct M-words. This prediction is borne out with complex gerunds in relative clauses. The upshot is that Uniqueness applies within M-words. In the next section, we turn to a different prediction of our analysis regarding allomorphy and derivational ordering in postsyntax.

5 Uniqueness meets allomorphy in the complex imperfective

In the preceding sections, we argued that Uniqueness blocks linking of the exponent of auxiliary agreement in Amharic when that exponent has already been linked to main verb agreement within the same M-word. This predicts that, if a distinct vocabulary entry were to be selected to realize main verb agreement, auxiliary agreement should resurface. In other words, our analysis predicts that low allomorphy of main verb agreement should *bleed* application of the Uniqueness constraint over Link at auxiliary agreement, thereby allowing erstwhile homophonous agreement markers to resurface. In this section, we show that this prediction is borne out in the complex imperfective in Amharic.

5.1 The interplay between Uniqueness and allomorphy

Imperfective verbs in Amharic are used in negative non-past main clauses and in all non-past subordinate clauses (Leslau 1995: 311). Imperfective verbs differ from gerunds (and from perfective verbs, see (16a)) in bearing *discontinuous* subject agreement, consisting of a prefix and, in certain φ -feature combinations, a suffix (Kramer 2023, To appear; see Harbour 2008, 2023; Hewett 2020, 2023a,b; and Shlonsky 2023 on discontinuous agreement in Afroasiatic). Roughly speaking, the prefix realizes person while the suffix realizes number and gender (see the aforementioned references for more details on the features realized by these affixes). The full paradigm of imperfective φ -agreement is given in Table 4.

	SG	PL
1	i -säbr	inni -säbr
2M	ti -säbr	ti -säbr- u
2F	ti -säbr- i	ti -säbr- u
3M	yi -säbr	yi -säbr- u
3F	ti -säbr	yi -säbr- u

Table 4: Simple imperfective of *säbbärä* ‘break’ (Leslau 1995: 301)

As we showed in (4), a simple imperfective verb can be combined with non-past *-allä* to form a complex imperfective verb, which is interpreted as present (progressive) or future. (40) presents an additional example.

- (40) gänzäb inni-fällig-allä-n (COMPLEX IPFV)
 money 1PL-want.IPFV-AUX.NPST-1PL
 ‘We want money.’ (Leslau 1995: 344)

When we look at the entire paradigm of complex imperfective agreement in Table 5, we once again find a disappearing agreement pattern, though different from what we observed with complex gerunds. While auxiliary agreement is overtly manifested throughout, the main verb agreement suffix disappears in the second and third person plural cells.

	SG	PL
1	i -säbr[-allä- h^w]	inni -säbr[-allä- n]
2M	ti -säbr[-allä- h]	ti -säbr- [-allä- atfɨihu]
2F	ti -säbr- i^y [-allä- f]	ti -säbr- [-allä- atfɨihu]
3M	yi -säbr[-allä- ä]	yi -säbr- [-allä- u]
3F	ti -säbr[-allä- ätɨɨ]	yi -säbr- [-allä- u]

Table 5: Complex imperfective of *säbbärä* ‘break’ (adapted from Leslau 1995: 342)

However, unlike with the complex gerund, disappearing agreement in the complex imperfective cannot be attributed to an anti-homophony pressure like Uniqueness. The only suffix missing in Table 5 is *-u* (cf. Table 4), and this exponent disappears regardless of the form of auxiliary agreement. Thus, in the second person plural form in (41), the expected main verb agreement suffix is *-u* (set in gray text) but auxiliary agreement is *-atfɨihu*.¹⁷

- (41) **ti**-säbr- u -allä- atfɨihu (> *tisäbrallatfɨuh*) (COMPLEX IPFV)
2-break.IPFV-PL -AUX.NPST-2PL
‘you (PL) break, will break’ (adapted from Leslau 1995: 342)

Uniqueness is thus a poor fit to account for the facts in Table 5. We contend that the relevant generalization has to do with the disappearance of a particular *exponent*: plural *-u*.

We propose (*pace* Kramer To appear: sec. 4.1) that the apparent disappearance of plural *-u* in the complex imperfective is contextual allomorphy resulting in a *-∅* exponent.¹⁸ Specifically, we propose the vocabulary entries in (42) for plural agreement.

¹⁷The 2PL imperfective agreement suffix is pronounced *-atfɨuh* word-finally (Leslau 1995: 288, 342).

¹⁸Alternatively, it might seem plausible for plural *-u* to be deleted to avoid vowel hiatus with the initial *a* of the auxiliary. However, as noted in Kramer (To appear: sec. 4.2), hiatus resulting from an /ua/ sequence is typically resolved in Amharic through glide insertion (not deletion). Additionally, a hiatus avoidance account has nothing to say about why auxiliary agreement is missing whenever material intervenes between the plural suffix and the auxiliary (see below). We therefore follow Kramer (To appear) in not pursuing a phonological explanation here (compare the phonological account of a certain kind of null prefix in Tigre in Faust 2015).

(42) *Vocabulary entries for plural agreement*

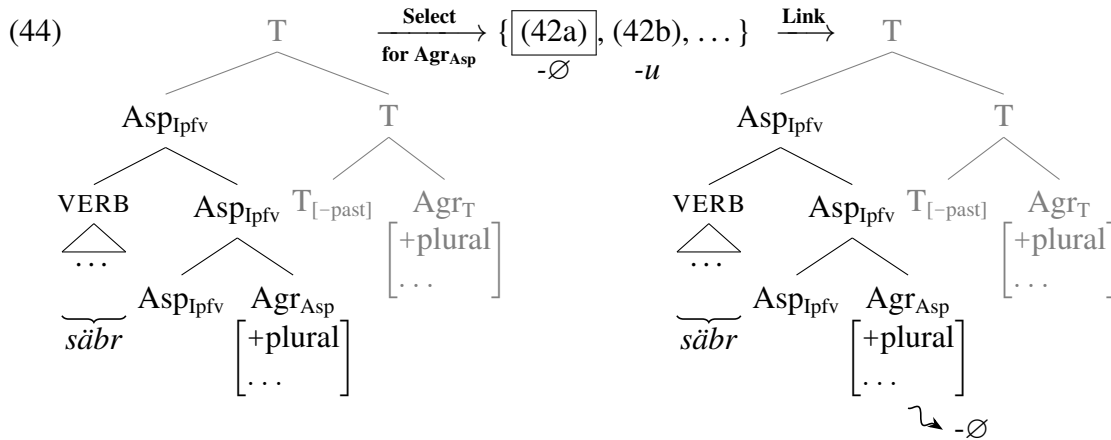
- a. $[Agr +plural] \rightarrow -\emptyset / Asp_{Ipfv} \text{ ____ } T_{[-past]}$
- b. $[Agr +plural] \rightarrow -u$

The more specific entry in (42a) requires that main verb Agr be linearly flanked by Asp_{Ipfv} on its left and by $T_{[-past]}$ on its right (more on this below),¹⁹ while (42b) is context-free.

Let us consider now how plural agreement suffixes are realized in the third person plural complex imperfective in (43) (repeated from (7) but with a $-\emptyset$ suffix; on deriving agreement prefixes in Semitic, see Hewett 2020, 2023a,b and Kramer 2023).

- (43) yi-säbr- $\emptyset$ -allä- u ($> yisäbrallu$) (COMPLEX IPFV)
 3-break.IPFV-PL -AUX.NPST-PL
 ‘they break, will break’ (adapted from Leslau 1995: 342)

The input to Vocabulary Insertion will be a complex head in T containing the main verb and the auxiliary, both with their own Agr morphemes bearing third person plural features. Vocabulary Insertion will apply cyclically, bottom up until it reaches Agr_{Asp} . Select will choose the highly specific vocabulary entry in (42a) to realize Agr_{Asp} per the Elsewhere Condition and Link will pair Agr_{Asp} with the exponent $-\emptyset$.



¹⁹The contextual specification in (42a) provides further evidence in favor of Hewett's (2023a) claim that discontinuous agreement affixes in Semitic always exhibit allomorphy that is sensitive to their linear position.

-*mm*. We discuss each in turn.²⁰ Object clitics attach at the right edge of the main verb, following main verb agreement suffixes but crucially preceding the auxiliary -*allä*. This is shown in (47) with a second person plural complex imperfective verb and the third person masculine singular object clitic -*t* (which is underlined for salience).

- (47) **ti-nägr-u** -t -allä-**atf̣f̣ihu** (> *tinägrutallatf̣f̣uh*) (COMPLEX IPFV)
 2-tell.IPFV-PL -3MSG.O -AUX.NPST-2PL
 ‘you all tell him’

Because the main verb agreement suffix is not contiguous with the non-past auxiliary, it is realized as -*u* instead of -∅—just as we would expect given the pair of vocabulary entries in (42) (compare (47) with (41)). As (48) shows, an intervening object clitic also causes plural -*u* to reappear on the main verb in third person plural forms (contrast (43)), again as we would expect. Most relevantly, the appearance of plural -*u* on the main verb in (48) coincides with the disappearance of plural -*u* from the auxiliary.

- (48) **yi-nägr-u** -t -allä □ (> *yinägrutall*) (COMPLEX IPFV)
 3-tell.IPFV-PL -3MSG.O -AUX.NPST
 ‘they tell him’ (adapted from Leslau 1995: 422)

This is *precisely* what we expect from a Uniqueness-based perspective: because -*u*, rather than -∅, was linked to Agr_{Asp} on the main verb, -*u* cannot also be linked to Agr_T.

We find a similar interaction with the focus marker -*mm*, whose placement is more flexible than the object clitics. This focus marker can appear either between the main verb and the auxiliary ((49)), parallel to object clitics, or to the immediate right of the auxiliary ((50)). As the following pair of examples illustrates, only when the focus marker intervenes between the verb and the auxiliary is the main verb plural suffix realized as -*u*, with the result that auxiliary agreement goes missing.

²⁰Previous work has discussed the interaction between agreement and object clitics/the focus marker in Amharic complex verbs; see e.g. Kramer (To appear) and the references cited in footnote 11 of that paper.

- (49) **yi-nägr-u** -mm -allä $\square$ (> *yinägrummall*) (COMPLEX IPFV)
3-tell.IPFV-PL -FOC -AUX.NPST
 ‘and they will tell’ (adapted from Leslau 1995: 882)
- (50) **yi-nägr-Ø** -allä-u-mm (> *yinägrallumm*) (COMPLEX IPFV)
3-tell.IPFV-PL -AUX.NPST-**PL-FOC**
 ‘they will tell’ (*ibid.*)

In other words, plural *-u* on the main verb and plural *-u* on the auxiliary are in complementary distribution when the two form an M-word, consonant with the predictions of our analysis. In summary, the distribution of plural *-u* in Amharic complex imperfectives is predicted by a combination of the Uniqueness constraint and adjacency-based allomorphy.

6 Interim summary

Let us recap our results thus far. Our analysis accounts for the disappearance of agreement in Amharic complex verbs with two postsyntactic tools. On the one hand, auxiliary agreement in Amharic disappears whenever it would be homophonous with a main verb agreement marker in the same complex head ((51)). This is borne out in the complex gerund in (52) and in certain third person plural complex imperfective verbs like (53).

- (51) AUX AGREEMENT GENERALIZATION IN AMHARIC COMPLEX VERBS
 Auxiliary agreement disappears whenever it would be homophonous with main verb agreement.
- (52) **därsä-n** -allä (*-n) (> *därsänall*) (COMPLEX GERUND)
 arrive.GER-**1PL** -AUX.NPST (*-**1PL**)
 ‘we have arrived’
- (53) **yi-nägr-u** -t -allä (*-u) (> *yinägrutall*) (COMPLEX IPFV)
3-tell.IPFV-PL -3MSG.O -AUX.NPST (*-**PL**)
 ‘they tell him’ (adapted from Leslau 1995: 422)

We accounted for disappearing homophonous agreement with the Uniqueness constraint in (54), which applies to the second step in a two-step Vocabulary Insertion process and whose domain is the M-word.

(54) **The Uniqueness constraint (on Link)**

In an M-word, the exponent associated with a vocabulary entry α is Linked no more than once.

On the other hand, we showed that main verb agreement in the complex imperfective is missing only when it is left-adjacent to the non-past auxiliary ((55)), as in (56).

(55) MAIN VERB SUFFIX GENERALIZATION IN AMHARIC COMPLEX IMPERFECTIVES
The plural suffix *-u* on lexical verbs in complex imperfectives is realized as a null allomorph whenever the lexical verb is adjacent to the non-past auxiliary.

(56) **y***i*-nägr-Ø -allä-**u** (> *yinägrallu*) (COMPLEX IPFV)
3-tell.IPFV-**PL** -AUX.NPST-**PL**
'they will tell' (adapted from Leslau 1995: 882)

Given that homophony plays no role in governing the distribution of main verb agreement, we argued that the (dis)appearance of the plural suffix is actually contextual allomorphy constrained by linear adjacency. Indeed, because Vocabulary Insertion applies bottom up and because main verb agreement is lower than auxiliary agreement, our analysis predicts that the form of auxiliary agreement should never determine the form of main verb agreement. This conclusion converges with claims that there is no outward-looking phonologically conditioned allomorphy (see e.g. Bobaljik 2000, Paster 2006, Embick 2010: esp. 62–63, Kalin 2020, 2022, Gouskova and Bobaljik To appear: sec. 4.1, and Kalin and Weisser To appear: sec. 3.1; but see Deal and Wolf 2017 for a different perspective).

In the next section, we show that each of the main components of our analysis—the Uniqueness constraint and zero allomorphy—boasts independent support from similar patterns of disappearing agreement in other Transversal South Ethiosemitic languages.

7 Variation across Ethiosemitic supports Uniqueness and allomorphy

Amharic is not alone in having complex, single M-word verbs that contain both an inflected main verb and an agreeing auxiliary. The other Ethiosemitic languages that use

complex verbs in this way are Argobba, Harari, Silt’e, Wolane, and Zay—the last three forming Eastern Gurage, a cluster of languages spoken in the Gurage Zone in southern Ethiopia (see Meyer 2011, 2019: 227). Together with Amharic, these languages comprise the Transversal South sub-branch of Ethiosemitic (Hetzron 1972: 36; Girma Awgichew Demeke 2001: 70, 80–81; Bulakh 2014). However, in the northern Ethiosemitic language Tigrinya, complex verbs consisting of an inflected main verb and an agreeing auxiliary do not form a single M-word. All else being equal, then, we might expect the Uniqueness constraint and zero allomorphy to regulate the distribution of agreement markers in the complex verbs of Transversal South Ethiosemitic languages, just as they do in Amharic, but to not do so in Tigrinya.

In this section, we show that these expectations are borne out, though in parameterized fashion in the Transversal South Ethiosemitic languages. The Uniqueness constraint is active in Eastern Gurage but inactive in Argobba and in non-standard Amharic. Similarly, zero allomorphy of certain main verb agreement suffixes is attested in Argobba but absent from Eastern Gurage and non-standard Amharic. If we take Uniqueness and zero allomorphy to be binary (independent) properties of a language, then, as Table 6 illustrates, all four possible language types are attested in Transversal South Ethiosemitic.

	[+Uniqueness]	[–Uniqueness]
[+zero allomorphy]	(standard) Amharic (sections 2–5)	Argobba (section 7.2)
[–zero allomorphy]	Eastern Gurage (section 7.1)	non-standard Amharic (section 7.3)

Table 6: Variation in Uniqueness and zero allomorphy predicts four language types across Transversal South Ethiosemitic

The fact that languages might vary in having zero allomorphy of agreement markers is unsurprising: this is the kind of low-level variation that we expect to find for memorized objects like vocabulary entries. What is perhaps more surprising is the discovery that the

Uniqueness constraint is subject to parametric variation. In other words, Uniqueness is not a necessary property of Vocabulary Insertion or of morphological systems in general. We make sense of this finding in section 7.4 by arguing that Uniqueness is a learnable, morphological constraint, similar to many other postsyntactic constraints and rules (e.g. those responsible for impoverishment, fission, fusion, etc).

We turn to Tigrinya in Section 7.5. Because periphrastic tenses do *not* form an M-word in Tigrinya, Uniqueness is irrelevant and we expect that homophonous agreement on the verb and the auxiliary will freely surface. We also predict that the agreement markers on the main verb will not have zero allomorphy triggered by the auxiliary.²¹ We show that both predictions are borne out, further widening the empirical reach of our account and re-confirming the M-word locality restriction on Uniqueness.²²

7.1 Eastern Gurage: [+Uniqueness, –zero allomorphy]

The Eastern Gurage languages Wolane, Silt’e, and Zay make use of complex imperfective verbs (though not complex gerunds), where the effects of Uniqueness are visible. For simplicity, we focus on Wolane in what follows.²³

Similar to Amharic, Wolane complex imperfective verbs (Table 9) combine a simple imperfective verb (Table 7) with the non-past auxiliary $-\bar{a}(n)$,²⁴ whose expected agreement

²¹Assuming that the locality restriction on contextual allomorphy is (at most) the M-word. See e.g., Bobaljik (2012) and Choi and Harley (2019).

²²See Sellami (2024: 288–294) for converging arguments for an M-word restriction on Uniqueness coming from their analysis of the distribution of agreement markers in Tunisian Arabic periphrastic vs. non-periphrastic tense constructions.

²³For the relevant data in the other languages, see the following sources: Silt’e (complex imperfective: Gutt 1997: 523; expected agreement on AUX: Gutt 1997: 521) and Zay (complex imperfective: Meyer 2005: 159–160, Table 32; expected agreement on AUX: Meyer 2005: 94–95, Table 14).

²⁴The precise factors which determine whether the Wolane auxiliary is realized as $-\bar{a}$ or as $-\bar{a}n$ are or-

markers are given in Table 8.²⁵

	SG	PL
1	y-sɛbr	y-sɛbr-nɛ
2M	t-sɛbr	t-sɛbr-u
2F	t-sɛbr-i	t-sɛbr-u
3M	y-sɛbr	y-sɛbr-u
3F	t-sɛbr	y-sɛbr-u

Table 7: Wolane simple imperfective of $\sqrt{\text{sbr}}$ ‘break’ (Meyer 2006: 110, Table 18)

	PL
1	-h^w -nɛ
2M	-hɛ -h^wm
2F	-ɟ -h^wm
3M	-∅ -u
3F	-t -u

Table 8: Wolane expected agreement on AUX (adapted from Meyer 2006: 108, Table 15)

	SG	PL
1	y-sɛbr[-ā-h^w]	y-sɛbr-nɛ[-ān □]
2M	t-sɛbr[-ā-hɛ]	t-sɛbr-u[-ā-h^wm]
2F	t-sɛbr-i[-ā-ɟ]	t-sɛbr-u[-ā-h^wm]
3M	y-sɛbr[-ān-∅]	y-sɛbr-u[-ān □]
3F	t-sɛbr[-ā-t]	y-sɛbr-u[-ān □]

Table 9: Wolane complex imperfective of $\sqrt{\text{sbr}}$ ‘break’ (adapted from Meyer 2006: 97)

thogonal to our discussion. We analyze this alternation as a morphophonological one, requiring reference to prosodic constituency—in particular, prosodic boundaries: the auxiliary is realized as $-\bar{a}n$ when it is at the right edge of a prosodic unit of some kind (e.g. that containing the complex imperfective in Table 9) and it is realized as $-\bar{a}$ when it is non-final within this domain. Importantly, we do not analyze the $-\bar{a}\sim-\bar{a}n$ alternation as arising via competition at Vocabulary Insertion.

²⁵We provide only underlying forms for Wolane data; see Meyer (2006) for corresponding surface forms.

Two lines of evidence support the claim that the non-past auxiliary *-ān* and the main verb form a single complex head in Wolane. First, the relative complementizer *ye-* in relative clauses attaches to the left of the entire complex verb (unlike in Amharic, see section 4), suggesting that the verb, the non-past auxiliary in T, and C all form a single complex head:

- (57) [CP **yε-ʔa-y-sɛbr-u** -y -ān] gəz
 [CP **C-NEG-3-break.IPFV-PL** -3MSG.O -AUX.NPST] thing
 ‘the thing which they do not break’ (Wolane; Meyer 2006: 127)

Second, *-ān* behaves phonologically like an enclitic attached to the main verb (Meyer 2006: 96), including participating in word-level phonological processes like vowel metathesis (p. 97). Non-past *-ān* contrasts with the past tense auxiliary *nɛr~nɛbɛr*, which does not fuse with the main verb (p. 54) and thus likely forms its own independent M-word.

There are two noteworthy aspects of the paradigm in Table 9. First, unlike in Amharic, main verb agreement is always overt in Wolane and never allomorphically realized as *-Ø*: there is no difference in main verb agreement between the simple imperfective paradigm in Table 7 and the complex imperfective paradigm in Table 9. Zero allomorphy thus does not play a role in determining the form of main verb agreement in Wolane.

Second, as in Amharic, Wolane auxiliary agreement vanishes when it would be homophonous with main verb agreement—the hallmark of Uniqueness. Compare the forms of expected auxiliary agreement in Table 8 with the complex imperfective paradigm in Table 9. The absence of third plural *-u* from the auxiliary parallels what we saw in Amharic (see (48)–(49)). The absence of first plural *-nɛ* from the auxiliary, on the other hand, is different from Amharic (cf. Table 5), but this is only because first plural imperfective agreement in Amharic is a monomorphemic prefix (cf. Table 4). First plural imperfective agreement in Wolane is crucially discontinuous (see Hewett 2023b: sec. 4 for further discussion and analysis) and utilizes the same suffix as the auxiliary—namely, *-nɛ*. We can account for disappearing auxiliary agreement in Wolane if Uniqueness is active in this lan-

guage: an exponent linked to main verb Agr will be unable to be relinked within the same M-word. It is a strength of our analysis that it automatically explains why innovative first plural auxiliary agreement also disappears in Wolane complex imperfectives.

7.2 Argobba: [–Uniqueness, +zero allomorphy]

The reverse state of affairs holds in the variety of Argobba spoken in the villages of Shonke and T’ollaha (see Girma Awgichew Demeke 2015 for an overview of Argobba varieties and additional references). Argobba, like Amharic, has both complex gerunds and complex imperfectives containing a non-past auxiliary *-(ha)ll* (Wetter 2010: 205–206). The available evidence suggests that the non-past auxiliary and main verb form a single complex head when they co-occur: the non-past auxiliary is consistently written as a single unit with the main verb and forms a single phonological word with it (Wetter 2010: 205). The non-past auxiliary can be contrasted with the past tense auxiliary *imber*, which is both orthographically and prosodically separated from the main verb (Wetter 2010: 208). Thus, *imber* likely forms a separate M-word from the main verb, so we set it aside henceforth.

Unlike in Amharic and Eastern Gurage, auxiliary agreement in Argobba non-past complex verbs is always overt (modulo the null realization of third masculine singular agreement: *-hall-Ø*). We focus on agreement in the complex gerund paradigm in Table 10, where main verb agreement and auxiliary agreement are homophonous in at least all second person forms (the two sets of agreement markers are fully distinct in the complex imperfective, see Wetter 2010: 400, rendering Uniqueness moot).

In order to derive forms like the second person plural *sebirr-xum-ll-xum* ‘you (pl.) have broken,’ an exponent like *-xum* (2PL) must be able to be linked multiple times within a single complex gerund M-word during Vocabulary Insertion. This indicates that the Uniqueness constraint must be *inactive* in Argobba.

	SG	PL
1	sɛbɪrr- e [-ll- ɛw]	sɛbɪrr- ɛna [-ll- na]
2M	sɛbɪrr- x [-ll- x]	sɛbɪrr- xum [-ll- xum]
2F	sɛbɪrr- f [-ll- f]	sɛbɪrr- xum [-ll- xum]
3M	sɛbɪrr- o [-hall- ∅]	sɛbɪrr- ɛm [-ll- ɛy]
3F	sɛbɪrr- a [-ll- ɛtfɪf]	sɛbɪrr- ɛm [-ll- ɛy]

Table 10: Argobba complex gerund of $\sqrt{\text{sbr}}$ ‘break’ (underlying forms; adapted from Wetter 2010: 401)

At the same time, certain main verb agreement suffixes are not always overt. As described and analyzed in Hewett (2023a: 1133–1136), the third person plural main verb suffix in Argobba (complex) imperfective verbs is allomorphically realized as $-\emptyset$ when it immediately precedes a direct object clitic ((58a)) and it is realized as $-u$ elsewhere ((58b)).

(58) *Argobba zero allomorphy of 3PL suffixes*

- a. **y-** awid $-\emptyset$ -**yyɛm** -ll -ɛy (>yawidiyyemilɛy)
3- tell.IPFV -**3PL** -**3PL.O** -AUX -3PL
‘they tell them’ (Wetter 2010: 394)
- b. **y-** awd $-u$ -ll -ɛy (>yawdulɛy)
3- tell.IPFV -**3PL** -AUX -3PL
‘they tell’ (Wetter 2010: 171)

The alternation between $-\emptyset$ and $-u$ is straightforwardly accounted for by competition between the vocabulary entries in (59) at Vocabulary Insertion (see Hewett 2023a: 1134–1136 for arguments that the disappearance of the third plural suffix is not due to postsyntactic *Obliteration*, on which see Arregi and Nevins 2007, 2012).

(59) *Vocabulary entries for 3PL complex imperfective suffixal agreement in Argobba*

- a. $\left[\begin{array}{l} -\text{participant} \\ +\text{plural} \end{array} \right] \rightarrow -\emptyset / \text{Asp}_{\text{IPFV}} \text{ ___ } \text{D}_{\text{OBJ}}$
- b. $\left[\begin{array}{l} -\text{participant} \\ +\text{plural} \end{array} \right] \rightarrow -u / \text{Asp}_{\text{IPFV}} \text{ ___ }$

To recap, in Argobba, Uniqueness must be inactive, accounting for the persistent appearance of homophonous agreement in complex verbs, while zero allomorphy accounts

for the absence of an overt third plural suffix before object clitics in complex imperfectives.

7.3 Non-standard Amharic: [–Uniqueness, –zero allomorphy]

The final parametric possibility predicted by our account involves a language without any disappearing agreement whatsoever. We argue that this possibility is instantiated by non-standard Amharic, which we take to include modern non-standard varieties described in reference grammars from the last century (Leslau 1995) in addition to Old Amharic, as attested in texts from before the mid-19th century (see Getatchew Haile 1983: 163; Appleyard 2003a: 234, 2003b: 115). As the following examples illustrate, homophonous agreement is attested in both complex gerunds and complex imperfectives in non-standard Amharic (see footnote 8 on the homophony of *-w* and *-u*).²⁶

- (60) tä-zägtä-w -allä-u (COMPLEX GERUND)
 NACT-close.GER-PL -AUX.NPST-PL
 ‘they have been closed’ (Old Amharic; Appleyard 2003a: 234)
- (61) yi-säbr-u -allä-u (> *yisäbruallu*) (COMPLEX IMPERFECTIVE)
 3-break.IPFV-PL -AUX.NPST-PL
 ‘they break, will break’ (Non-standard Amharic; adapted from Leslau 1995: 342;
 see also Goldenberg 1977: 494)

To account for the appearance of homophonous auxiliary agreement in non-standard Amharic, the Uniqueness constraint must be inactive. This will allow an exponent like plural *-u* to be linked more than once within an M-word (i.e. to Agr_{Asp} and to Agr_T). Moreover, to account for the presence of plural *-u* on the imperfective main verb in (61), we propose that there is not a more specific zero allomorph available to realize the plural suffix when it is left-adjacent to the non-past auxiliary, unlike in modern standard Amharic (cf. (42a)). Without Uniqueness or zero allomorphy to suppress agreement, double subject marking is

²⁶We assume that the main verb and auxiliary form a single complex head in non-standard Amharic, which is in line with the available orthographic evidence (see e.g., Getatchew Haile 1983: 163).

the norm in non-standard Amharic complex verbs, even when the exponents are identical.

7.4 Variation in (anti-)homophony suggests Uniqueness is learnable

In this section so far, we have argued for the independent plausibility of the two post-syntactic tools that we used to derive disappearing agreement in Amharic: the Uniqueness constraint and zero allomorphy. Nevertheless, the variation documented so far highlights an important additional conclusion—namely, that the Uniqueness constraint is not a necessary property of Vocabulary Insertion. While Uniqueness is active in (modern standard) Amharic and Eastern Gurage, it is patently not active in Argobba and non-standard Amharic. Rather than being exceptional, we contend that this is compatible with—and in fact, fully expected by—our morphological analysis of (anti-)homophony effects. Previous work has shown that morphological microvariation can be accounted for by positing fine-grained differences in the inventory of postsyntactic constraints (or ‘filters’) and rules across dialects/idiolects. Thus, variation in the (in)activity of Uniqueness parallels garden variety variation in constraints on feature coexponence that give rise to a range of attested syncretisms and morphological discontinuities. The latter type of variation is commonly assumed in work on impoverishment and fission (see e.g. Noyer 1992, Arregi and Nevins 2018: 637–638, esp. fn. 14, and Hewett 2023a: 1102–1103, esp. fn. 7, 2023b).²⁷ The upshot is that variation in morphological constraints like Uniqueness is wholly unsurprising.

Still, while Uniqueness might be an arbitrary (but undeniable) part of the grammars of certain Ethiosemitic languages, its existence sheds important light on the architecture of postsyntax. As emphasized in sections 3.2–3.3, Uniqueness must only apply to Link—the second step in a two-step Vocabulary Insertion process. Restricting Uniqueness in this way

²⁷See also Arregi and Nevins (2012: ch. 5, 2018, 2022) on using morphotactic constraints and repairs to account for variation in patterns of morpheme displacement (e.g. metathesis vs. doubling) across languages.

is necessary to ensure that it suppresses the realization of homophonous auxiliary agreement without triggering a retreat to an elsewhere exponent. If this reasoning is sound, then the existence of Uniqueness supports the idea that ‘Vocabulary Insertion’ is not a unitary operation. It must consist of at least two sub-operations or sub-routines—one responsible for allomorph selection (i.e. blocking) and one responsible for pairing morphosyntactic terminals with morphophonological forms. The results of this paper are thus novel and offer significant support for a key development in our understanding of Vocabulary Insertion—arguably the most central process in a realizational morphological theory like DM.

7.5 Tigrinya: No M-Words

In the Transversal South Ethiosemitic languages discussed so far in this section, all of the complex tenses have formed M-words. However, the situation is different in the Northern Ethiosemitic language Tigrinya. Specifically, Tigrinya complex tenses do *not* form M-words, and we show in this subsection that, as predicted by our analysis, there are accordingly no effects of Uniqueness or zero allomorphy in Tigrinya.

Like Amharic, Tigrinya has complex gerunds and imperfectives, both of which are formed with a non-past auxiliary (on Tigrinya complex verbs, see e.g., Voigt 1977; Tsehay Teferra 1979; Tesfay Tewolde Yohannes 2016). Unlike Amharic, Tigrinya has *two* non-past auxiliaries: *?allo*, an existential copula likely cognate with Amharic *allä*, and *?iyyu*, the equative copula. A gerund can combine with either auxiliary ((62)–(63)) with a present perfect interpretation (Tsehay Teferra 1979: 120, Tesfay Tewolde Yohannes 2016: 204).²⁸

- (62) binyam siwwa säty-u **?all-o** (COMPLEX GERUND)
 Binyam.M local.beer drink.GER-3MSG **AUX.NPST-3MSG**
 ‘Binyam has drunk local beer’ (Tigrinya; Tesfay Tewolde Yohannes 2016: 205)

²⁸The gerund combined with *?iyyu* has an additional interpretation of simple past, according to Tesfay Tewolde Yohannes (2016: 206).

- (63) binyam siwwa säty-u **?iyy-u** (COMPLEX GERUND)
 Binyam.M local.beer drink.GER-3MSG **AUX.NPST-3MSG**
 ‘Binyam has drunk local beer’ (Tigrinya; Tesfay Tewolde Yohannes 2016: 205)

An imperfective main verb can also combine with either auxiliary, albeit with different interpretations for each. When an imperfective verb combines with *?allo*, the result is present progressive, as in (64).

- (64) säb yi-zäwwir **?all-o** (COMPLEX IMPERFECTIVE)
 man 3-IPFV.walk **AUX.NPST-3MSG**
 ‘A man is walking.’ (Tigrinya; Tesfay Tewolde Yohannes 2016: 197)

When an imperfective verb combines with *?iyyu*, the possible interpretations are somewhat harder to pin down, but they likely include habitual and/or modal meanings ((65); see Tsehaye Teferra 1979: 125, Tesfay Tewolde Yohannes 2016: 197-198).

- (65) säb yi-zäwwir **?iyy-u** (COMPLEX IMPERFECTIVE)
 man 3-IPFV.walk **AUX.NPST-3MSG**
 ‘A man can walk/walks.’ (Tigrinya; Tesfay Tewolde Yohannes 2016: 198)

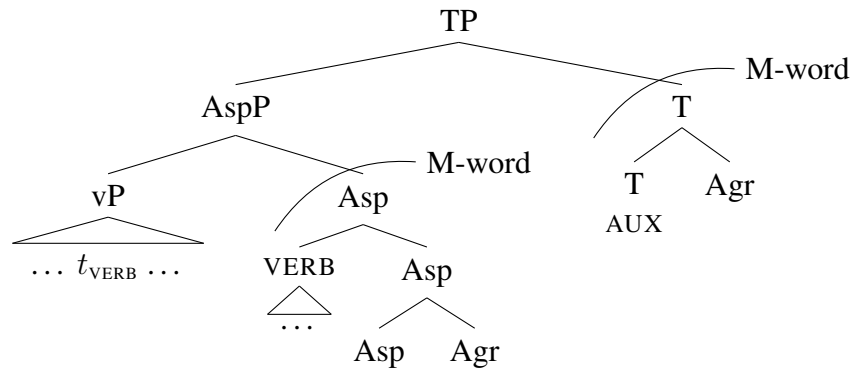
As in the other Ethiosemitic languages discussed so far, complex verbs in Tigrinya have subject agreement markers on both the main verb and the auxiliary. For example, in (62), the gerund main verb has a third masculine singular suffix *-u* and the auxiliary has a third masculine singular suffix *-o*.²⁹

We now turn to the M-word status of Tigrinya complex verbs. In much (if not all) previous work on Tigrinya complex verbal syntax (see e.g., Diertani 2011; Tesfay Tewolde Yohannes 2016; Cacchioli 2023), the main verb and the auxiliary have been analyzed as

²⁹Gerunds and imperfective verbs also agree with the subject in simple verb clauses without auxiliaries (see e.g., Tsehaye Teferra 1979: 118,122). Moreover, both *?allo* and *?iyyu* agree with the subject when they are used as copular main verbs (see e.g., Bulakh 2019: 193-194). We therefore treat Tigrinya complex main verbs as containing two phi probes, one for the main verb and for the auxiliary, as we have assumed for all the Ethiosemitic complex verbs in this paper.

two separate M-Words. Auxiliaries express tense distinctions in Tigrinya (see e.g., Tesfay Tewolde Yohannes 2016), and thus the structure of a Tigrinya complex verb has been understood roughly to be as in (66) where the auxiliary is in T and the main verb remains below T (although it might raise to a lower verbal projection like Aspect, as in (66)).

- (66) *Basic structure of Tigrinya complex verbs* (based on Diertani 2011: 49, (3.37); see also Tesfay Tewolde Yohannes 2016: 200, (14))



There are several pieces of evidence that support treating the main verb and auxiliary as distinct M-words. First, they are written as two separate orthographic units in Tigrinya (see e.g., Mulugeta Girmay Melles 2001: 54, 61), unlike in the Transversal South Ethiosemitic languages where they are written as a single unit.³⁰ Second, the auxiliary *?iyyu* immediately precedes the main verb in certain information structure contexts (see Cacchioli 2024: (42)). Under Cacchioli (2024)’s analysis of these clauses, the auxiliary has moved to a higher position than the verb. If the auxiliary can move separately from the verb, then the auxiliary and the main verb must be separate M-words because excorporation from a complex head/M-word is impossible (see Baker 1988: 73 and Embick and Noyer 2001: 575:fn. 25). Finally, and most tellingly, unrelated material can intervene between the auxiliary and the main verb in certain instances of auxiliary movement. For example, in (67), the subject *Kidane* intervenes between the auxiliary and the main verb (both bolded).

³⁰Tigrinya uses the same writing system as the Transversal South Ethiosemitic languages: an abugida-type of syllabary known as the *fidäl*.

- (67) *nī mǎn ʔiyy-ä kidane räʔay-ä ʔil-u*
 ACC who AUX.NPST-1SG Kidane.M see.GER-1SG COMP-3MSG
 ‘Who did Kidane say that he saw?’ (Tigrinya; Spadine 2020: 30)

(67) raises many additional questions, some of which have been answered and some of which remain to be addressed. Spadine (2020) argues extensively that *ʔilu* is a complementizer, that this is a monoclausal construction (the English translation does not reflect the structure), and that the first-person singular agreement reflects a shifted indexical. However, she does not speculate on the auxiliary movement; it may be related to the focus constructions that trigger auxiliary movement for Cacchioli (2024). In any case, the precise analysis of the auxiliary movement can be set aside for future work. For our purposes, the most important property of (67) is that the auxiliary again moves independently from the main verb, resulting in additional material intervening between them, and thus the auxiliary and the main verb cannot be an M-word.

Since the main verb and the non-past auxiliary do not form an M-word in Tigrinya, the Uniqueness constraint is irrelevant. Therefore, homophonous subject agreement on the main verb and non-past auxiliary should surface freely. This prediction is borne out, as already seen with the auxiliary *ʔiyyu* in (63), with homophonous *-u* for third masculine singular and in (67), with homophonous *ä* for 1st singular. Homophonous verb-auxiliary agreement also occurs with the auxiliary *ʔallo*, e.g., in the third feminine singular, (68).

- (68) *mäs‘i-a ʔall-a* (COMPLEX GERUND)
 come.GER-3FSG AUX.NPST-3FSG
 ‘She has come’ (Tigrinya; Diertani and Eilam 2010: 3)

Thus, Tigrinya complex verbs provide additional support for the M-word locality restriction on the Uniqueness constraint.³¹

³¹We focus on complex gerunds to make this point because imperfective main verbs have prefixal/discontinuous agreement; it would take us too far field to demonstrate that some of the imperfective

Additionally, assuming that contextual allomorphy is limited to (at most) within M-words (see e.g., Choi and Harley 2019), then Tigrinya complex imperfectives are predicted *not* to display zero allomorphy. The auxiliary will be too far away to condition any contextual allomorphy of the agreement marker on the main verb. This prediction is borne out as well; for example, third masculine plural imperfective verbs take a *-u* suffix, as in (69).

- (69) *yî-nägr-u* (SIMPLE IMPERFECTIVE)
 3-tell.IPFV-PL
 ‘They(m.) come’ (Tigrinya; Bulakh 2019: 187)

The same exponent is present on the main verb in the complex imperfective, as in (70).

- (70) *yî-s‘awät-u* ?all-äwu (COMPLEX IMPERFECTIVE)
 3-play.IPFV-PL AUX.NPST-3PL
 ‘They(m.) are playing’ (Tigrinya; Diertani and Eilam 2010: 5)

Thus, Tigrinya does not display zero allomorphy on complex imperfective main verbs, as predicted by our account because the verb and auxiliary are not in the same M-word.

Recall from Section 7.4 that languages in which complex verbs are M-words vary in whether they show effects of Uniqueness and zero allomorphy. However, Tigrinya is a distinct case in that it presents strong evidence that the main verb and auxiliary do not form an M-word. Under our analysis, such a language *must* lack effects of Uniqueness and zero allomorphy, and that is what we have demonstrated in this section.

Overall in Section 7, we have shown that the Uniqueness constraint and zero allomorphy shape agreement paradigms in closely related Ethiosemitic languages just as our analysis would predict. We interpret the enhanced empirical coverage of our analysis as providing strong support in favor of the Uniqueness-based approach and we demonstrate in Section 8 that alternative approaches are less empirically successful.

agreement markers are the same vocabulary entries as the auxiliary agreement markers. See Hewett (2023a) on the derivation of imperfective agreement marking across Semitic.

8 Comparison with alternative approaches

Up to this point, we have focused on developing the Uniqueness Constraint-based analysis of agreement disappearance in Ethiosemitic. However, there are some alternative approaches to consider, and we discuss them in this section, including an approach based on morphological movement (section 8.1) and approaches based on featural identity (section 8.2).³² To the best of our knowledge, Kramer (To appear) and Diertani and Eilam (2010) are the only previous formal approaches to Ethiosemitic complex verbs; in addition to discussing these proposals, we also include alternative approaches to similar data in other languages (e.g., Kinyalolo’s Constraint for Bantu languages).

8.1 Local Dislocation

Kramer (To appear) develops a Distributed Morphology analysis of the disappearing subject agreement in the complex imperfective in Amharic. Specifically, she proposes that the plural suffix *-u* undergoes Local Dislocation (Embick and Noyer 2001) when it is immediately left adjacent to *allä*. For example, to derive (71), the *-u* suffix starts out as a third person agreement suffix on the verb; at PF, it undergoes Local Dislocation and moves to the right of the auxiliary.

- (71) *yi-säbr* *-allä-u* (> *yisäbrallu*) (COMPLEX IPFV)
 3-break.IPFV -AUX.NPST-PL
 ‘they break, will break’ (adapted from Leslau 1995: 342)

Because the plural agreement suffix *-u* and the auxiliary *allä* must be adjacent, two key facts about disappearing subject agreement are simultaneously explained: (i) *-u* “re-appears” when any material intervenes between it and the auxiliary (explained in our account be-

³²See the appendix for arguments against analyzing the Ethiosemitic data with a theory like Morphology as Syntax (Collins and Kayne 2023) which builds words in the syntax but which adopts early insertion.

cause of the locality conditions on the allomorphy of plural agreement), (ii) the auxiliary lacks *-u* in intervention contexts (explained in our account via Uniqueness).

However, a Local Dislocation account has some disadvantages. Because it is tailored for the distribution of main verb agreement, it does not make any predictions about the distribution of auxiliary agreement. It is thus forced to take on certain additional assumptions in order to ensure the right form of agreement on the auxiliary (e.g., that the auxiliary lacks a third plural agreement suffix underlyingly and that there is a haplology operation deleting the Locally-Dislocated main verb suffix in certain contexts). In contrast, on our approach, the distribution of auxiliary agreement is directly explained by the Uniqueness constraint.

Additionally, a Local Dislocation approach does not extend easily to the complex gerund. For example, it incorrectly predicts Local Dislocation in the third plural forms where the main verb suffix and the auxiliary are adjacent.

- (72) *Local Dislocation analysis of complex gerund makes incorrect prediction*
 säbrä-w-all ~> *säbrä-all-u
 break.GER-PL-AUX.NPST
 ‘they have broken’

Therefore, we conclude that the Uniqueness approach is more promising than a Local Dislocation analysis.

8.2 Feature Identity

Uniqueness regulates homophony in form, not identity of features. Could a feature-based approach fare equally well (or better) in explaining Ethiosemitic complex verb agreement? In this section, we answer this question negatively, demonstrating how approaches to disappearing agreement in Ethiosemitic based on feature identity do not hold up to close scrutiny. We begin by showing that the feature-based approach developed for Amharic in Diertani and Eilam (2010) and Diertani (2011) is too powerful and lacks a principled account of

directionality (i.e., which agreement marker 'disappears'). We then discuss Kinyalolo's Constraint, a classic constraint on multiple agreement marking in Bantu. We argue that, while Kinyalolo's Constraint is not an adequate account of Ethiosemitic agreement, it is likely part of the same family of morphological constraints on repetition as Uniqueness.

8.2.1 The Morphological Anti-Redundancy Condition

In Diertani and Eilam (2010) and Diertani (2011), disappearing agreement in Amharic complex verbs is explained via a 'Uniqueness'-like constraint called the Morphological Anti-Redundancy Condition:

(73) **Morphological Anti-Redundancy Condition (MARC)**

Speakers will delete a morpheme when they assume its feature content is identical to that of another morpheme within the same Morphological-word. (Diertani and Eilam 2010: 4)

(73) is like Uniqueness in that it regulates when identical elements can co-occur within an M-word; however, it is different from Uniqueness in that the relevant identity relation is stated over morphosyntactic features (and not vocabulary entries). As a result, (73) does not make the right predictions without significant additional stipulations.³³ Specifically, if deletion is triggered by feature identity, then agreement will disappear in every cell of the paradigm. Moreover, (73) says nothing about directionality, and hence it fails to predict that it is always *higher* (homophonous) agreement that disappears. Kramer (To appear) also points out that (73) does not explain the 'intervention effects' of the complex imperfective, i.e., why the plural main verb agreement suffix is overt when it is not adjacent to

³³These stipulations include a requirement that agreement be morphologically realized on Asp in the complex gerund but C in the complex imperfective and that the non-homophonous 1sg and 3sg agreement suffixes retained on the main verb gerund express different features than their counterparts on the auxiliary (there is no independent evidence that the suffixes on the gerund express different features).

the auxiliary. Overall, then the Uniqueness constraint argued for here makes more accurate predictions than (73) with fewer additional unmotivated assumptions.

8.2.2 Kinyalolo’s Constraint

While Uniqueness bans the Linking of identical *vocabulary entries*, Kinyalolo’s Constraint, originally developed for Bantu languages, bans identical *morphosyntactic features*. We will show that Kinyalolo’s Constraint does not explain Ethiosemitic complex verb agreement and speculate on the relationship between Kinyalolo’s Constraint and Uniqueness.

In the Bantu language Kilega, multiple heads in the verbal functional spine agree with the subject; thus, in at least some instances, a single internally-complex ‘word’ is predicted to contain multiple subject agreement markers (Kinyalolo 1991; Carstens 2005).³⁴ However, as noted by Kinyalolo and Carstens, this prediction is not borne out. Instead, there is only one subject agreement marker per word. For example, in (74), there are three verbal words and each has a subject agreement marker that agrees in noun class 4 with the subject.

- (74) Mikoko **z-á-bézág-á zǐ-se z-á-sínz-u-a**
 4.sheep 4-A-be-FV 4-about.to 4-ASP2-slaughter-PASS-FV
 ‘Sheep were about to be slaughtered’ (Kilega; Carstens 2005: 254, (52a))

In (75), there are two verbal words: ‘be still’ and the main verb ‘cook.’ Accordingly, there are only two agreement markers, both agreeing in noun class 6 with the subject.

- (75) Masungá **má-kili m-á-yik-u-á**
 6.yam 6-be.still 6-ASP2-cook-PASS-FV
 ‘The yams are still being cooked’ (Kilega; Carstens 2005: 253, (12a))

³⁴It is not entirely clear whether the verbal ‘words’ in Kilega are M-words; the syntactic, morphological, and phonological formation of verbal words in Bantu languages is complex and not obvious from orthographic boundaries (see Pietraszko 2018 for relevant discussion). However, for the sake of simplicity and for ease of comparison with Uniqueness, we tentatively assume that Kinyalolo’s Constraint holds over M-words.

To capture this generalization, Kinyalolo (1991) proposed a constraint on the morphology of agreement that Carstens later dubbed ‘Kinyalolo’s Constraint’ (henceforth, KC).

(76) **Kinyalolo’s Constraint (KC)**

In an adjoined [head:MH/RK], AGR on a lower head is silent iff its features are predictable from AGR on a higher head. (modified from Carstens 2003: 407)

KC ensures that, in an adjoined head, agreement features on a lower head are not pronounced if they are “predictable from” (usually read as ‘identical to’) agreement features on a higher head. Evidence from complementizer agreement in Kilega supports the idea that it is agreement on the *lower* head that is not pronounced: Carstens (2005) argues that certain M-words would contain both complementizer agreement on C and subject agreement on T, but, in these cases, only the higher, complementizer agreement is pronounced.

KC is similar to Uniqueness in that both are constraints on morphological repetition. However, KC cannot account for complex verb agreement in Ethiosemitic. First, it is too powerful; it is stated over features (like the Morphological Anti-Redundancy Condition; see Section 8.2.1), so it incorrectly predicts that complex verbs with two agreement markers are impossible. Second, the directionality requirement is not upheld: in Amharic, when there is suppression of an agreement marker in the complex gerund, it is the *lower* agreement marker that is realized and the *higher* marker which is absent.

Yet, Uniqueness does not work well for the Kilega data in turn—the directionality is incorrect and it is potentially not powerful enough.³⁵ Also, KC has been argued to explain feature-sensitive deletion in a range of languages (see e.g., Baker 2010; Henderson 2011; Alok and Baker 2018; Alok 2021; Tyler and Kastner 2022; Oxford 2023; Jou 2024). If we interpret this body of work as support for KC, but also consider this paper as support for Uniqueness, then what is the relationship between KC and Uniqueness in the grammar?

³⁵Whether or not it is powerful enough will depend on how much homophony is found within the relevant subject agreement paradigms for verbal elements in Kilega.

Building on Nevins (2012)’s pioneering work on haplology, we speculate that there are two kinds of repetition-avoidance constraints that hold over M-words:³⁶

(77) **Constraints on Morphological Repetition within an M-word**

- a. A constraint that holds *before* Vocabulary Insertion, sensitive to feature identity and structural position (= KC, in part)
- b. A constraint that holds *during* Vocabulary Insertion, sensitive to homophony and order of insertion (= Uniqueness)

The two kinds of constraints differ in timing: (77a) holds before Vocabulary Insertion, so it cannot access morphophonological information, whereas (77b) holds during Vocabulary Insertion and can access morphophonological information. If we adopt the constraint-and-repair approach to PF operations in Arregi and Nevins (2012), then KC is a combination of (i) a (77a)-type constraint, and (ii) a PF operation which deletes an agreement node, e.g., Obliteration (effectively treating KC as an extreme kind of dissimilatory Impoverishment; see e.g., Arregi and Nevins 2012; Yuan 2020; Oxford 2023). In contrast, Uniqueness instantiates (77b), and the ‘repair’ is to not Link a homophonous vocabulary entry.

To the best of our knowledge, KC effects have not previously been treated as part of the (DM-focused) typology of repetition-avoidance operations, and we hope that future work will provide support for this approach.³⁷ One interesting prediction is that a single language could be subject to both KC and Uniqueness, manifesting different kinds of repetition avoidance with distinct empirical signatures; such a language may in fact be Amharic, where KC has been invoked to prevent object agreement in unaccusatives (see Baker 2012) and multiple object marking in clitic doubling structures (see Kramer 2014).

³⁶We leave for future work how to relate Uniqueness and KC to anti-homophony constraints that are sensitive to linear order (see e.g., Kramer 2023 for one such constraint in Amharic).

³⁷Intriguingly, both KC and Uniqueness might be parameterized such that languages vary in whether they are subject to (either of) these constraints. See Baker (2012: 60) and Baker and Kramer (2018: 1081, fn.46) on parameterizing KC and Section 7.4 on parameterizing Uniqueness.

Overall, it is clear that neither Local Dislocation nor feature-based approaches are the right explanation for Ethiosemitic complex verb agreement. However, KC and Uniqueness appear to both be constraints on morphological repetition that hold at different points of the postsyntactic derivation.

9 Conclusion

This paper began with a puzzle: why does subject agreement ‘disappear’ from certain complex verbs in Amharic? To solve this puzzle, we posited the anti-homophony constraint Uniqueness and demonstrated how the interplay of Uniqueness with allomorphy generates the intricate patterns of agreement in complex verbs not just in Amharic, but also across a swath of the Ethiosemitic language family. The analysis has major repercussions for our understanding of Vocabulary Insertion (which, we argue, is necessarily comprised of two steps) and it provides support for multiple key planks of current morphosyntactic theorizing, from bottom-up Vocabulary Insertion (which explains the directionality of agreement ‘disappearance’) to the important role of M-words (the domain over which the Uniqueness constraint applies). From a larger perspective, the results of this work will hopefully be used as a starting point for analyzing complex verb agreement across languages, and as a springboard towards investigating anti-homophony constraints in the grammar.

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A Apparent coordination with imperfective verbs in Amharic

Imperfective verbs are only licit in (affirmative) non-past main clauses in Amharic with the support of the non-past auxiliary (see Section 5 of the main text). Because of this requirement, certain coordinated clauses seem to have a non-past auxiliary scoping over two imperfective main verbs; otherwise the first imperfective verb would lack auxiliary support. An example of this kind of clause is in (78), which seems to include a non-past auxiliary in T scoping over conjoined imperfective AspPs.

- (78) [[gimmaʃ k'än y-ars]_{AspP} -inna [y-arf]_{AspP} -allä-ä]_{TP}
 [[half day 3-plough.IPFV]_{AspP} -CONJ [3-rest.IPFV]_{AspP} -AUX.NPST-3MSG]_{TP}
 ‘He ploughs half a day and then he rests’ (adapted from Leslau 1995: 315)

If the subject of the coordinated clauses is third plural, then the imperfective verb in the first conjunct has the plural *-u* suffix, whereas the imperfective verb in the second conjunct has the zero allomorph.

- (79) [[gimmaʃ k'än y-ars-**u**]_{AspP} -nna [y-arf-Ø]_{AspP} -allä-**u**]_{TP}
 [[half day 3-plough.IPFV-**PL**]_{AspP} -CONJ [3-rest.IPFV-PL]_{AspP} -AUX.NPST-**PL**]_{TP}
 ‘They plough half a day and then they rest’ (Kramer To appear)

(79) is compatible with our approach to complex imperfectives under at least two different analyses. On one analysis, the non-past auxiliary attaches to the second verb before Vocabulary Insertion, so that the auxiliary is linearly adjacent to the second verb and thus triggers insertion of the zero allomorph discussed in Section 5 of the main text. Under a second analysis, the clause structure of (79) does not involve coordination, despite initial appearances. Instead, the first imperfective verb is in an adjoined subordinate clause, likely with a causal interpretation. Imperfective verbs in embedded clauses do not require auxiliary support, so there would be no need for the auxiliary to be associated with this verb. The second imperfective verb would then be part of a complex imperfective along with the auxiliary in the main clause. (80) re-analyzes (79) under this subordination approach.

- (80) [[gimmaf k'än y-ars-**u**-*нна*]_{CP} y-arf-Ø-*allä-u*]_{CP}
 [[half day 3-plough.IPFV-**PL-C**]_{CP} 3-rest.IPFV-PL-AUX.NPST-**PL**]_{CP}
 'Because they ploughed half a day, they rest' (Kramer To appear)

In (80), the suffix/enclitic *-нна* is treated as a complementizer (not a conjunction).

There is some evidence in favor of a causal subordination analysis, and against a coordination analysis. For example, under a subordination analysis, the first verb and the main verb are predicted to be able to have different subjects, and this is attested (see e.g., Leslau 1995: 315). In contrast, under a coordination analysis, it would be much more difficult to account for different subjects; because AspPs are coordinated, the subjects would likely have to undergo ATB movement to Spec,TP and thus be forced to be identical.

Additional support for the idea that *-нна* is a complementizer (rather than a conjunction) comes from sequences of two complex imperfective verbs. Complex imperfective verbs can be either asyndetically coordinated or connected with *-нна*. When they are asyndetically coordinated, they are interpreted as simple coordination, as in (81).

- (81) k'än-u-n mulu yi-bäl-*allä-ä* yi-t'ät't'-*allä-ä*
 day-DEF-ACC whole 3-eat.IPFV-AUX.NPST-3MSG 3-drink.IPFV-AUX.NPST-3MSG
 'All day long he eats and drinks' (adapted from Leslau 1995: 344)

However, when they are connected with *-нна*, the first complex imperfective is interpreted as if it were in a causal adjoined clause (see also discussion of this interpretation in Hailu Fulass 1980; Girma A. Demeke 2012; Desalegn Asfawwesen 2016).

- (82) abbat-e yi-mät'-*allä-ä-нна* wändim-e
 father-my 3-come.IPFV-AUX.NPST-3MSG-C brother-my
 yi-t'äbbik'-äw-*allä-ä*
 3-wait.IPFV-3MSG.O-AUX.NPST-3MSG
 'Since my father is coming, my brother will wait for him' (adapted from Leslau 1995: 345)

The contrast in interpretation between (81) and (82) points towards *-нна* being a (subordinating) complementizer, and (true) coordination being accomplished by a null conjunction,

at least in these contexts.

Overall, the apparent ‘coordination’ facts in (79) are compatible with a zero allomorphy approach to complex imperfectives, especially given the evidence in favor of a subordination analysis of clauses with *-nna*.³⁸

B An argument against Morphology as Syntax

In the main text, we argue that understanding the Uniqueness constraint reveals deeper properties of Vocabulary Insertion. Despite this, the main text does not address alternative approaches which eschew Vocabulary Insertion altogether in favor of *early insertion*—the hypothesis that the lexical items manipulated by the syntax are specified with phonological forms. In this section, we consider one such alternative—namely, Collins and Kayne (2023). We argue that disappearing auxiliary agreement provides a novel argument against Collins and Kayne’s approach to word-building and in favor of late insertionist approaches like DM.

To begin, recall how our Uniqueness-based analysis accounts for the disappearance of homophonous auxiliary agreement in Amharic. Given a complex head with more than one terminal node in need of exponence, the Uniqueness constraint prevents a single rule from supplying an exponent to more than one terminal node within that complex head. Such a constraint is formulable in DM because exponents (and the rules of which they are the output) are separated from the terminal nodes they realize (in accordance with arguments by Beard 1966, Aronoff 1976, Anderson 1992, and Halle and Marantz 1993, among others); that is, DM adopts *late insertion* (see e.g. Halle and Marantz 1994: 275–276; Marantz

³⁸In this section, we focused on conjoining imperfective verbs; we leave for future work to determine whether complex gerunds are licit in structures like (80).

1995; Harley 2014; and Embick 2015: ch. 5, To appear).³⁹

Let us now turn to Collins and Kayne (2023), who develop an approach to morphology dubbed *Morphology as Syntax* (MaS) in which (i) ‘words’ are built by syntactic operations and (ii) “[t]here is no Late Insertion” (p. 17, (41)). According to Collins and Kayne, “Merge itself (not vocabulary insertion) introduces phonological material into syntactic structures” (p. 17). This is because syntax operates on (at least) lexical items (LIs), which are defined as follows:

- (83) LI: {FF, PHON}, (Collins and Kayne 2023: 3, (3))
where FF is a set of formal features and PHON is a sequence of phonological segments.

Because lexical items contain PHON, phonological forms are part of the *input* to syntactic computation.

This brings us to the question of how MaS might account for disappearing auxiliary agreement in Ethiosemitic. Collins and Kayne (2023: sec. 3) cite a number of syntactic principles which they argue license null morphemes in varying contexts. We will show that none of these principles are sufficient to account for disappearing auxiliary agreement in Ethiosemitic. This constitutes a novel empirical argument against MaS, as well as other approaches which assume both early insertion and syntactic word-building.

The first kind of null exponence discussed by Collins and Kayne arises under movement: if a constituent X undergoes movement, non-highest copies of X (in earlier terminology, ‘traces’ of X) typically go unpronounced. This kind of explanation strikes us as extremely ill-suited to account for the Ethiosemitic data. We are not aware of any evidence that the two agreement markers in Amharic complex verbs are linked via syntactic movement. Moreover, even if we were to posit movement between them, a movement analysis would wrongly lead us to expect that auxiliary agreement, being higher than main verb

³⁹On late insertion in the related theory *nanosyntax*, see Baunaz and Lander (2018: 12).

agreement, should be pronounced. For these reasons, we reject a movement analysis of missing auxiliary agreement.

The second route to null exponence mentioned by Collins and Kayne involves ellipsis: a constituent X can be unpronounced due to being elided if X has an identical antecedent Y. In order for an ellipsis analysis of missing auxiliary agreement in Ethiosemitic to get off the ground, the identity condition on ellipsis would need to be sensitive to the morphophonological *form* of agreement markers (and not their morphosyntactic or semantic features), since only homophonous agreement markers go missing (see section 2 of the main text). But the identity condition on ellipsis is not sensitive to form (see esp. Merchant 2015: 210ff.). Ellipsis too, then, is an analytical dead end for the Ethiosemitic facts.

A third possible way to derive null exponence according to Collins and Kayne is via a constraint that prevents a head and its specifier from being simultaneously overt (i.e. a more general form of the Doubly Filled Comp Filter). At least two facts militate against such an analysis for Ethiosemitic. First, the two agreement markers do not obviously form a Spec-Head relation; hence, there is no obvious way to make one's appearance dependent on the other under this approach. Second, a purely structural constraint against realizing both a head and its specifier should apply to *all* (auxiliary) agreement markers, irrespective of their form, contrary to fact (recall: only homophonous auxiliary agreement is missing).⁴⁰

A fourth type of null exponence admitted by Collins and Kayne follows from the Invisible Category Principle of Emonds (1987):

⁴⁰A similar line of reasoning rules out another potential culprit for missing auxiliary agreement in Amharic. Collins and Kayne cite Kayne (2006) for the proposal that specifiers of phases are generally null. Suppose for the sake of argument that Ethiosemitic auxiliary agreement occupies the specifier of a phase head. Still, under the reasonable assumption that the structural position of auxiliary agreement is unaffected by the form of main verb agreement, this analysis also fails to predict a role for homophony in determining the distribution of agreement markers.

(84) **Invisible Category Principle**

A closed category B with positively specified features C_i may remain empty throughout a syntactic derivation if the features C_i (save possibly B itself) are all alternatively realized in a phrasal sister of B.” (Emonds 1987: 615,

(5))

An account of missing auxiliary agreement in terms of the Invisible Category Principle faces a by-now familiar problem: because (84) only makes reference to (morphosyntactic) features, it cannot (or at least cannot easily) explain why only homophonous agreement markers disappear.

The remaining routes to null exponence discussed by Collins and Kayne rely on morphemes which are null in the lexicon (i.e. whose PHON in (83) is ‘ $\emptyset$ ’) and whose syntactic distribution is restricted in various ways. We contend that there is no straightforward way to account for missing auxiliary agreement in Ethiosemitic with such lexically null morphemes. While it is certainly possible to stipulate that auxiliary agreement is null with certain φ -feature combinations in the local context of the gerund,⁴¹ such an analysis would, just like previous accounts, completely miss the form-based generalization about *which* φ -feature combinations are null. This is because the disappearance of auxiliary agreement in Ethiosemitic follows a *meta* pattern, akin to metasyncretism (see Williams 1994; Bobaljik 2002; Frampton 2002; Harley 2008) and metafission (Hewett 2023a,b), which is not statable at the level of individual restrictions on morphemes.⁴²

As we have shown, the various syntactic conditions proposed by Collins and Kayne

⁴¹Though note that, if the distribution of allomorphs is regulated by syntactic selection in MaS (see Collins and Kayne 2023: 10, esp. (23)), and if a head H can only select for an XP that merges with H or with a projection of H (see e.g. Landau 2007: 488, (3a), Merchant 2019, and Zyman 2024; but see Hewett 2024 and references cited therein for arguments for a constrained form of non-local selection), then this would require rethinking the structure of complex verbs in Ethiosemitic. This is because, on our analysis, auxiliary agreement realizes Agr_T, which is not immediately local to Asp or Agr_{Asp}.

⁴²Note too that the contexts in which auxiliary agreement is null in Ethiosemitic do not form a well-defined

(2023) to license null exponents are not sensitive to form. These conditions are thus insufficient to account for the Ethiosemitic data presented in this paper. More generally, we suspect that *any* analysis that adopts early insertion and builds words in the syntax will face similar hurdles in accounting for attested anti-homophony effects in Ethiosemitic complex verbs. By contrast, we have shown that realizational theories like DM can account for the facts crucially because the rules assigning exponents to morphemes are distinct from the morphemes themselves: anti-homophony is accounted for by restricting the application of vocabulary entries at PF.

syntactic natural class; compare the instances of morphosyntactic haplology discussed in Nevins (2012: secs. 3.2–3.4), which pay attention to featural identity between two (or more) morphemes.